

Langkah bicara / persiapan — tidak perlu layar tertentu

► TAMPILKAN / LAKUKAN

- Buka aplikasi IOMS. Siapkan 3 akun: Admin (Rita), Sales (Beni), Warehouse (Wawan).
- Tips: buka 3 jendela / incognito berbeda supaya cepat berpindah peran tanpa logout-login terus.
- Pastikan data demo siap: ada Purchase Order berstatus **Ordered** (PO baru berstatus Draft dan hanya Admin yang bisa men-*submit*-nya) dan Sales Order yang bisa dibuat/disetujui.

🗣️ KATAKAN (baca naskah ini)

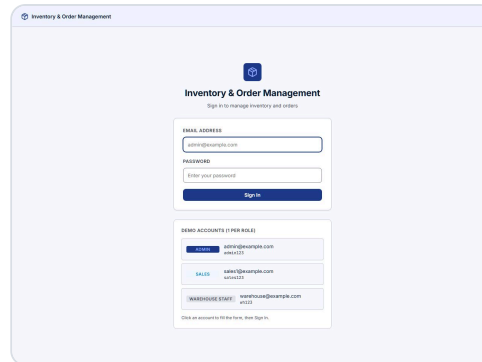
"Sebelum demo, perkenalan singkat: aplikasi ini mengelola stok dan pesanan untuk perusahaan dengan beberapa gudang. Ada tiga pengguna — Admin, Sales, dan Warehouse — dan saya akan mendemokan perjalanan sebuah pesanan dari dibuat sampai barangnya keluar, berpindah dari satu peran ke peran berikutnya."

💡 Kenapa 3 role dipisah — ucapkan di momen ini

Ini Segregation of Duties — supaya tidak ada satu orang yang menguasai seluruh proses (membuat sekaligus menyetujui pesannya sendiri). Kontrol anti-kecurangan klasik.

Peta dokumen ini

Halaman demo (langkah 1-11, plus sub-langkah "b/c/d/e" untuk tiap fitur Admin, Sales, Warehouse) diikuti lampiran: **A** Sesi, Redis & bcrypt · **B** Apa yang di-cache Memcached · **C/D/E** Checklist "how-to" per role (Admin, Sales, Warehouse) · **F** Koreksi terhadap versi sebelumnya.



Tampilan aplikasi yang harus Anda tampilkan

► TAMPILKAN / LAKUKAN

- Tampilkan halaman login. Masukkan email & password, klik Masuk.

🎤 KATAKAN (baca naskah ini)

"Pertama saya login. Saya ketik email dan password lalu Masuk."

💡 Selip alasan — Sesi & Redis — ucapkan di momen ini

"Saat login berhasil, identitas saya disimpan sebagai sesi. Sesi ini saya simpan di **Redis** — bukan di database utama — karena sesi butuh masa berlaku otomatis dan harus cepat dibaca di setiap halaman. Kalau Redis kebetulan mati, aplikasi otomatis beralih ke penyimpanan file, jadi login tetap jalan." • Keamanan: password dicek dengan **bcrypt**, tidak pernah disimpan polos.

Maksud kedua kalimat di atas (terverifikasi di kode)

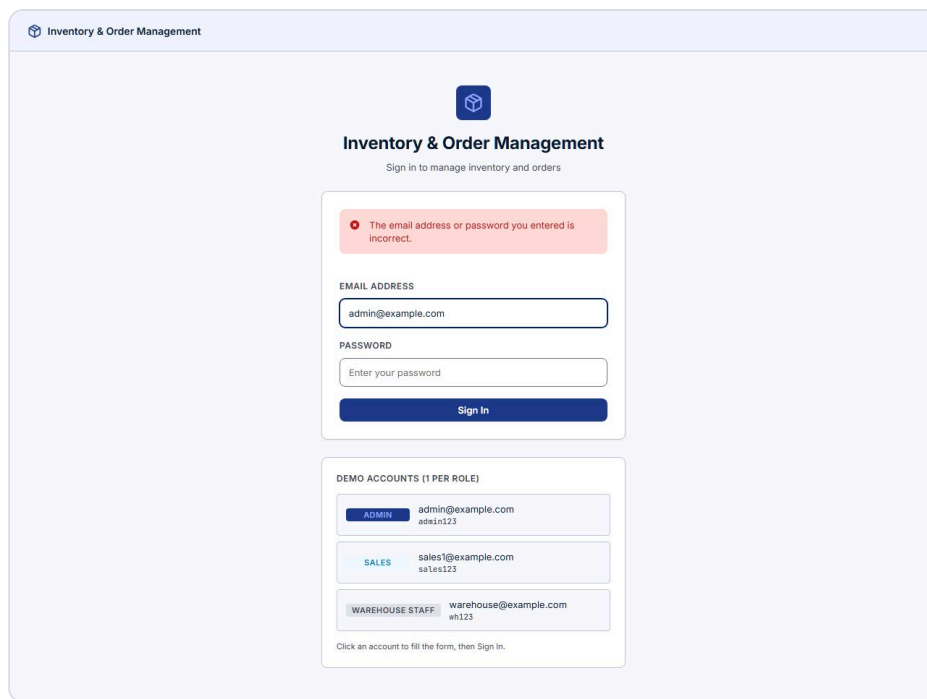
1. "Redis mati → beralih ke file, login tetap jalan"

- **Sesi** = data "siapa yang sedang login" (`user_id` , `role` , `last_activity`) di sisi server; browser hanya memegang cookie berisi ID acak. Normalnya disimpan di **Redis** (cepat, kedaluwarsa otomatis 3600 dtk, tidak membebani MySQL). Hanya sesi di Redis — bukan data bisnis/stok.
- Tiap request, `SessionManager` mengecek koneksi ke Redis (timeout 0,5 dtk). Gagal → tulis peringatan di log dan pakai **sesi berbasis file** (`session.save_handler=files`). Tanpa ini, Redis mati membuat semua request gagal/menggantung dan tak ada yang bisa login. Sudah diuji: dengan Redis palsu, sesi tetap aktif di mode file.
- **Batasan jujur (inferensi dari kode)**: sesi lama tidak ikut pindah, jadi pengguna yang sudah login harus login ulang saat Redis mati; login baru bisa, tapi sesinya tersimpan di file lokal.

2. "Password dicek dengan bcrypt, tidak pernah disimpan polos"

- Saat user dibuat/ganti password: `password_hash($pw, PASSWORD_BCRYPT)` → di DB hanya hash `$2y$10$...` (60 karakter, salt acak, sengaja lambat). Password asli tidak ada di DB, sesi, maupun log.
- Saat login: `password_verify($pw, hash)` — hash tidak bisa "dibalik"; dibandingkan, bukan didekripsi. Salah/akun nonaktif/email tak dikenal → pesan sama.

Versi lebih lengkap: Lampiran A.



The screenshot shows a web application titled "Inventory & Order Management" with a subtitle "Sign in to manage inventory and orders". The login form has two input fields: "EMAIL ADDRESS" with the value "admin@example.com" and "PASSWORD" with the placeholder "Enter your password". A red error message at the top of the form states: "The email address or password you entered is incorrect." Below the form is a "Sign In" button. At the bottom, there is a section for "DEMO ACCOUNTS (1 PER ROLE)" with three rows: "ADMIN" (admin@example.com, admin123), "SALES" (sales@example.com, sales123), and "WAREHOUSE STAFF" (warehouse@example.com, wh123). A note at the bottom says "Click an account to fill the form, then Sign In."

Tampilan aplikasi yang harus Anda tampilkan

► TAMPILKAN / LAKUKAN

- Ketik password salah → muncul pesan umum *"The email address or password you entered is incorrect."* (tidak menyebut mana yang salah).

🎤 KATAKAN (baca naskah ini)

"Kalau saya salah memasukkan password, pesannya sengaja dibuat umum — tidak memberi tahu mana yang salah."

💡 Kenapa pesan umum — ucapkan di momen ini

Supaya penyerang tidak bisa menebak email mana yang terdaftar (anti user-enumeration). Percobaan gagal juga dicatat untuk audit (event `login_failed` ; email yang dicoba sengaja tidak ikut disimpan di log). Akun nonaktif mendapat pesan yang sama persis.

Halaman login: tautan **Forgot your password?** di bawah tombol Sign In

Halaman Forgot password (publik, tidak perlu login)

Setelah submit: pesan umum yang sama

Email yang TIDAK terdaftar: jawabannya persis sama

► TAMPILKAN / LAKUKAN

- Di halaman login klik **Forgot your password?**, isi email (mis. `sales1@example.com`), klik **Send reset link**.
- Tampil pesan: *"If that email address belongs to an active account, we have queued a password reset email for it..."*
- Ulangi dengan email yang tidak ada → pesan identik.

🔊 KATAKAN (baca naskah ini)

"Kalau pengguna lupa password, dia minta link reset lewat email. Aplikasi hanya mencatat permintaan dan selalu menjawab dengan kalimat yang sama, baik emailnya terdaftar maupun tidak."

💡 Kenapa jawabannya selalu sama

Anti user-enumeration (sama seperti pesan login gagal): penyerang tidak bisa mengetes email mana yang terdaftar. Email yang tidak valid, tidak terdaftar, nonaktif, atau terkena rate limit

tetap dijawab sama (`PasswordResetService::requestReset`). Server hanya menulis satu baris di `password_reset_requests` dengan `status = 0` ; email tidak dikirim saat itu juga, melainkan oleh cron (langkah berikutnya).

Langkah 1d — Lupa password: bukti di database, cron, dan email

Tabel password_reset_attempts — setiap permintaan dicatat (email hanya sebagai hash) untuk rate limit 3email & 10IP per jendela waktu

id	ip_address	email_hash_prefix	created_at
9	127.0.0.1	d1e8a643913a11a3	2026-10-06 00:58:40
10	127.0.0.1	ea24083a240811e2	2026-10-06 00:58:41

Tabel `password_reset_attempts` : tiap permintaan dicatat (email hanya hash) untuk rate limit

Tabel password_reset_requests — sebelum cron (status 0 = email belum terkirim)

id	user_id	status	attempts	sent_at	expires_at	used_at
4	2	0	0	NULL	NULL	NULL

Tabel `password_reset_requests` — status 0 = menunggu dikirim; `token_hash` masih kosong

Menjalankan: php scripts/send-password-reset-emails.php

```
$ php scripts/send-password-reset-emails.php
[2026-10-06 00:58:43 UTC] password-reset emails: sent=1 failed=0 skipped=0 expired=0
```

Cron `send-password-reset-emails.php` : `sent=1`

Tabel password_reset_requests — setelah cron (status 1 = terkirim; token hanya disimpan sebagai hash)

id	user_id	status	attempts	sent_at	expires_at	used_at	token_hash_prefix
4	2	1	1	2026-10-06 00:58:43	2026-10-06 01:28:43	NULL	cc14b2043c3779c

Setelah cron: status 1, token tersimpan sebagai HASH, `expires_at` = 30 menit dari kirim

Email yang dikirim (MAIL_TRANSPORT=file → storage/mail/*.eml)

```
Date: Tue, 06 Oct 2026 00:58:43 +0000
From: Inventory & Order Management <noreply@ioms.local>
To: Beni <sales1@example.com>
Subject: Reset your password
Message-ID: <edcd3c86ab97273345f79f642da3ea54@localhost>
MIME-Version: 1.0
Content-Type: text/plain; charset=UTF-8
Content-Transfer-Encoding: base64

Hello Beni,

We received a request to reset the password of your Inventory & Order Management account.
Open the link below to choose a new password. It works once and expires in 30 minutes:

http://127.0.0.1:8090/reset-password#token=f817897aae3c408086bf5998d9844a6a7e1cd9007e7714a409b93d9cfdc752ba

If you did not ask for this, you can ignore this email; your password stays unchanged.
```

Email yang dikirim (di lingkungan demo lokal ditulis ke `storage/mail/*.eml` ; di server memakai SMTP). Link memakai `#token=...`

► TAMPILKAN / LAKUKAN

- (Opsional, bila ditanya penguji) Perlihatkan baris `status = 0` lalu jalankan `docker compose exec cron php scripts/send-password-reset-emails.php` (di produksi cron berjalan otomatis tiap 3 menit) → baris menjadi `status = 1`.
- Buka email: ada link sekali pakai.

"Pengiriman email dikerjakan proses cron terpisah, jadi permintaan di web cepat dan tidak membocorkan apa pun. Token acak 64 karakter hex hanya dikirim lewat email; di database hanya tersimpan hash-nya."

💡 Detail keamanan (terverifikasi di kode & README)

Token = `bin2hex(random_bytes(32))` , disimpan sebagai `sha256` ; berlaku **30 menit** sejak dikirim dan **sekali pakai**. Rate limit **3 permintaan/email** dan **10/IP** per 60 menit. Satu permintaan aktif per user. Pengiriman gagal diulang dengan jeda 3, 6, 9... menit sampai 5 percobaan lalu `status = 2` ; permintaan yang tidak terkirim dalam 60 menit ditinggalkan (`status = 2`). Klaim baris memakai UPDATE atomik sehingga tidak ada email ganda. Token ditaruh di *fragment* URL (`#token=`) — browser tidak mengirim fragment ke server, sehingga token tidak masuk access log; halaman reset membacanya lewat JavaScript (`views/auth/reset-password.php`). Bahwa access log tidak merekamnya adalah inferensi dari perilaku browser.

Halaman reset (dibuka dari link email)

Password < 6 karakter ditolak

Konfirmasi tidak sama ditolak

Form terisi benar (min. 6 karakter, sama)

Sukses: kembali ke login dengan pesan hijau

Login dengan password baru berhasil



Choose a new password

Use at least 6 characters.

This password reset link is invalid or has expired. Please request a new one.

NEW PASSWORD

CONFIRM NEW PASSWORD

Change password

Back to sign in

Tabel password_reset_requests — setelah dipakai (used_at terisi, token hangus)

id	user_id	status	sent_at	expires_at	used_at
4	2	1	2026-10-06 00:58:43	2026-10-06 01:28:43	2026-10-06 00:58:48

Tabel setelah dipakai: `used_at` terisi

Link yang sama dipakai lagi: ditolak (invalid/expired)

► TAMPILKAN / LAKUKAN

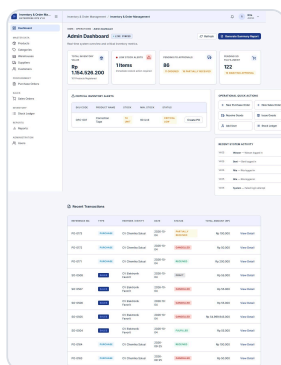
- Buka link dari email. Coba password pendek (ditolak), lalu konfirmasi berbeda (ditolak), lalu password valid di kedua kolom → **Change password**.
- Login memakai password baru → masuk dashboard.
- Buka lagi link yang sama → ditolak: *"This password reset link is invalid or has expired. Please request a new one."*

🎤 KATAKAN

"Link ini hanya bisa dipakai sekali dan kedaluwarsa dalam 30 menit. Setelah dipakai, link yang sama otomatis tidak berlaku, dan password baru disimpan sebagai bcrypt, sama seperti saat admin membuat user."

💡 Catatan

Password baru di-hash dengan `PASSWORD_BCRYPT` (`password_hash`), minimal 6 karakter (aturan yang sama dengan `UserService`). Hasilnya dicatat di event log sebagai `password_reset` (tanpa password/token). **Setelah demo ini pulihkan password akun demo** (mis. `sales1@example.com` kembali ke `sales123`) lewat Admin → Users → Edit, agar langkah berikutnya tetap memakai kredensial di halaman Persiapan.



Tampilan aplikasi yang harus Anda tampilkan

► TAMPILKAN / LAKUKAN

- Tampilkan dashboard Admin: ringkasan stok, pesanan, aktivitas.
- Tunjuk menu kiri LENGKAP: Dashboard, Master Data (Products, Categories, Warehouses, Suppliers, Customers), Procurement, Sales, Inventory, Reports, Administration (Users).

🎤 KATAKAN (baca naskah ini)

"Saya masuk sebagai Admin. Dashboard menampilkan ringkasan kondisi gudang dan pesanan. Perhatikan menu di kiri — Admin punya akses penuh ke semua modul."

💡 Selip alasan — Role & otorisasi — ucapkan di momen ini

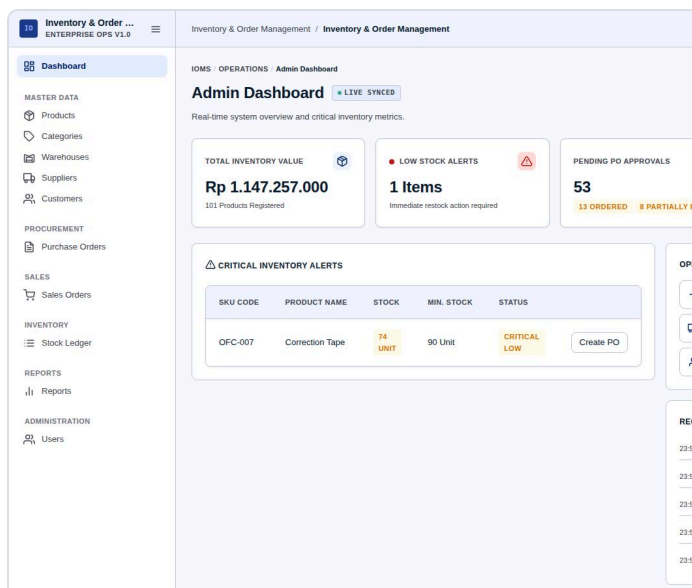
"Menu yang muncul ditentukan izin role saya. Tapi menyembunyikan menu saja bukan keamanan — setiap halaman tetap dicek di server. Nanti saya tunjukkan kalau role lain memaksa buka menu terlarang, dia tetap ditolak."

Soal cache Memcached — data apa tepatnya?

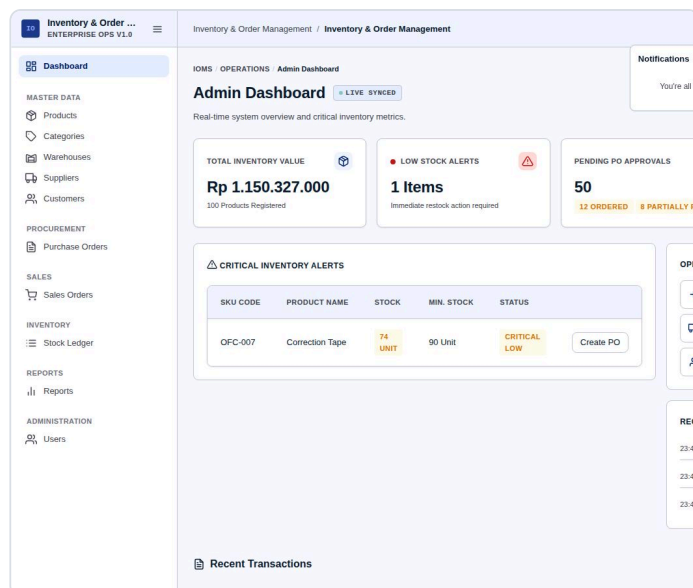
Angka dashboard (nilai inventori, low stock, jumlah PO/SO, antrean) dibaca langsung dari MySQL tiap halaman dibuka (badge "LIVE SYNCED") — **tidak di-cache**. Yang di-cache **Memcached** hanya dua hal (izin role dipindah ke Redis, lihat bawah tabel):

Key	Isi	TTL	Invalidasi
product:<SKU>	Satu produk per SKU (dipakai API <code>/api/products/{sku}/availability</code>); bukan stok	5 menit	Dihapus saat produk diubah / dinonaktifkan / diaktifkan
file_validation.rules	Aturan tipe gambar upload produk	1 jam	Hanya kedaluwarsa TTL

Stok tidak pernah di-cache (selalu MySQL + `SELECT ... FOR UPDATE`) agar anti-oversell tak bergantung data usang. Bila Memcached mati, aplikasi tetap jalan penuh dari MySQL. Tabel izin role (`role_permissions.grouped`, 1 jam) kini di-cache di **Redis** (database terpisah dari sesi), bukan Memcached. Versi lengkap: Lampiran B.



Dashboard Admin: kartu ringkasan, Critical Inventory Alerts (Create PO), Operational Quick Actions



Bel notifikasi di header (Mark All as Read bila ada peringatan)

► TAMPILKAN / LAKUKAN

- Klik **Refresh** → angka dimuat ulang (badge "LIVE SYNCED").
- Tunjuk empat kartu: **Total Inventory Value** (+ jumlah produk terdaftar), **Low Stock Alerts**, **Pending PO Approvals** (rincian per status), **Pending SO Fulfillment** (badge "n Awaiting Approval" bila ada).
- Di panel **Critical Inventory Alerts**, klik **Create PO** pada satu produk → terbuka form Purchase Order (kembali dengan Back/menu).
- Tunjuk **Recent Transactions** (5 PO + 5 SO terbaru, tombol View Detail) dan **Recent System Activity** (5 aktivitas terakhir: waktu — nama user — aktivitas).
- Tunjuk **Operational Quick Actions**: New Purchase Order, New Sales Order, Receive Goods, Issue Goods, Add User, Stock Ledger.
- Klik **bel notifikasi** di header → daftar peringatan stok rendah; klik **Mark All as Read** → hitungan jadi 0. Tombol **Generate Summary Report** membuka halaman Reports.

🔊 KATAKAN (baca naskah ini)

"Dashboard ini bukan sekadar tampilan. Dari sini Admin langsung melihat apa yang perlu ditindak: stok yang menipis bisa langsung dibuatkan Purchase Order, pesanan yang menunggu persetujuan terlihat jelas, dan di bawahnya ada jejak aktivitas terakhir untuk audit."

💡 Selip alasan — Dashboard sebagai pusat tindakan — ucapkan di momen ini

"Semua angka ini dibaca langsung dari database setiap halaman dibuka, jadi selalu mutakhir. Notifikasi stok rendah dibuat oleh job terjadwal yang berjalan tiap 15 menit, dan tidak diduplikasi selama notifikasi sebelumnya belum dibaca."

SKU CODE	PRODUCT NAME	UOM	PRICE (Rp)	TOTAL PHYSICAL STOCK	STATUS
QTC-001	A4 Paper 80mm Office Supplies	sheet	Rp 55,000	282 sheet	IN STOCK
QSB-023	Acrylic Sheet 2mm Raw Materials	sheet	Rp 145,000	80 sheet	IN STOCK
SPR-024	Air Filter Regulator Spare Parts	pcs	Rp 105,000	88 pcs	IN STOCK
QSB-002	Aluminum Rod 1x Raw Materials	pcs	Rp 140,000	80 pcs	IN STOCK
QSB-008	Aluminum Sheet 2mm Raw Materials	sheet	Rp 295,000	90 sheet	IN STOCK
SPR-012	Ball Bearing 6005 Spare Parts	pcs	Rp 48,000	80 pcs	IN STOCK
QTC-002	Ballpoint Pen Box Office Supplies	box	Rp 40,000	80 box	IN STOCK
SPR-003	Bearing 6004 Spare Parts	pcs	Rp 42,000	77 pcs	IN STOCK
QTC-009	Binder Clip 100mm Box Office Supplies	box	Rp 22,000	80 box	IN STOCK
ELEC-004	Bluetooth Speaker Electronics	pcs	Rp 299,000	80 pcs	IN STOCK

Tampilan aplikasi yang harus Anda tampilkan

► TAMPILKAN / LAKUKAN

- Buka menu Products (Master Data). Tunjuk daftar produk + ringkasan stok.
- Jelaskan: master data (produk, kategori, gudang, supplier, customer) dikelola Admin — dipakai semua role.

🎤 KATAKAN (baca naskah ini)

"Sebelum ada transaksi, Admin menyiapkan data induk: produk, gudang, supplier, dan customer. Data inilah yang dipakai Sales dan Warehouse nanti."

💡 Selip alasan — Database & soft-delete — ucapkan di momen ini

"Semua data ini tersimpan di MySQL — satu sumber kebenaran. Saya pilih database relasional karena datanya saling terhubung dan butuh transaksi. Dan kalau data induk 'dihapus', sebenarnya hanya dinonaktifkan, bukan hilang — supaya pesanan lama yang merujuknya tidak rusak." (Pengecualian terverifikasi: Kategori punya aksi hapus sungguhan, tetapi ditolak bila masih dipakai produk.)

Products

Manage catalog items, category classifications, and master inventory units.

Summary:

- TOTAL PRODUCTS (SKU): 100 SKU
- NORMAL STOCK ITEMS: 99 SKU
- LOW STOCK WARNING: 1 SKU
- OUT OF STOCK: 0 SKU

Filters:

- Sort Products: Name A-Z
- Product Name: [Search]
- Category: [Select category]
- Stock Status: All Active
- Warehouse: [Select warehouse]

SKU CODE	PRODUCT NAME	UOM	PRICE (RP)	TOTAL PHYSICAL STOCK	STATUS
OFC-001	A4 Paper Ream Office Supplies	ream	Rp 55.000	280 ream	IN STOCK
SA-020	Acrylic Sheet 3mm Raw Material	sheet	Rp 140.000	80 sheet	IN STOCK

Daftar Products + KPI dan filter

Create Product

Basic Information

- SKU CODE: OFC-007
- Barcode / EAN: [Please Enter The Barcode Number Here...]
- Product Name: Produk Demo Duplikat
- Description: [Please Enter The Product Description Here. Include Physical Specifications, Storage Conditions, And Handling Instructions...]

Status & Categorization

- Category: Electronics
- Unit of Measure: Pcs
- Product Status: ☒ Active

Initial Stock & Notes

- Warehouse Location: [No initial allocation]
- Initial Quantity: 0

Form Add Product terisi

Add New Product

Changes could not be saved
This SKU is already in use.

Basic Information

- SKU CODE: OFC-007
- Barcode / EAN: [Please Enter The Barcode Number Here...]
- Product Name: Produk Demo Duplikat
- Description: [Please Enter The Product Description Here. Include Physical Specifications, Storage Conditions, And Handling Instructions...]

Status & Categorization

- Category: Electronics
- Unit of Measure: Pcs
- Product Status: ☒ Active

Initial Stock & Notes

- Warehouse Location: [Please Select Warehouse Location]

SKU kembar ditolak

Products

Manage catalog items, category classifications, and master inventory units.

Summary:

- TOTAL PRODUCTS (SKU): 101 SKU
- NORMAL STOCK ITEMS: 99 SKU
- LOW STOCK WARNING: 1 SKU
- OUT OF STOCK: 1 SKU

Filters:

- Sort Products: Name A-Z
- Product Name: [Search]
- Category: [Select category]
- Stock Status: All Active
- Warehouse: [Select warehouse]

SKU CODE	PRODUCT NAME	UOM	PRICE (RP)	TOTAL PHYSICAL STOCK	STATUS
OFC-001	A4 Paper Ream Office Supplies	ream	Rp 55.000	280 ream	IN STOCK
SA-020	Acrylic Sheet 3mm	sheet	Rp 140.000	80 sheet	IN STOCK

Produk baru tersimpan

Edit Product

Basic Information

- SKU CODE: DEMO-001
- Barcode / EAN: 8991234567890
- Product Name: Stapler Demo Mini
- Description: Produk contoh untuk demo

Status & Categorization

- Category: Electronics
- Unit of Measure: Pcs
- Product Status: ☒ Active

Pricing & Inventory Control

Reset Form Save Product

Edit produk (ubah harga jual)

Products

Manage catalog items, category classifications, and master inventory units.

Summary:

- TOTAL PRODUCTS (SKU): 100 SKU
- NORMAL STOCK ITEMS: 99 SKU
- LOW STOCK WARNING: 1 SKU
- OUT OF STOCK: 0 SKU

Filters:

- Sort Products: Name A-Z
- Product Name: [Search]
- Category: [Select category]
- Stock Status: All Active
- Warehouse: [Select warehouse]

No products match your search or filter criteria.

Setelah Deactivate: hilang dari daftar default

► TAMPILKAN / LAKUKAN

- Products → **Add Product**. Isi: SKU (unik), barcode (opsional), nama (min. 3 huruf), deskripsi, kategori, unit, harga beli, harga jual, reorder point, status aktif; opsional gudang & qty awal dan gambar (JPG/PNG/WebP). Simpan → produk muncul di daftar.
- Tunjukkan validasi**: isi SKU yang sudah ada → ditolak dengan pesan; unggah file bukan gambar → ditolak.
- Cari/filter di daftar: SKU, nama, kategori, gudang, status stok; ubah urutan dan jumlah baris per halaman.
- Buka detail produk → lihat stok per gudang & pergerakan terakhir. Klik **Edit** → ubah harga atau reorder point → Simpan.
- Klik **Deactivate** → produk hilang dari daftar default dan dari dropdown PO/SO baru; tampilkan lewat filter status *Inactive*, lalu **Activate** untuk mengembalikan.

KATAKAN (baca naskah ini)

"Admin bisa menambah produk baru lengkap dengan harga, batas minimum stok, dan gambarnya. Kalau SKU kembar, sistem menolak. Kalau saya ubah harga atau batas minimum, tinggal Edit. Dan kalau produk sudah tidak dijual, saya nonaktifkan — bukan dihapus."

Selip alasan — Validasi & nonaktif — ucapkan di momen ini

"Produk tidak pernah dihapus permanen karena sudah terhubung ke riwayat PO/SO; dinonaktifkan agar histori tetap utuh. Upload gambar dicek ekstensi dan isi file (byte signature) — bukan sekadar nama file — dan gambar dikonversi ke WebP."

Categories

Organize product catalog classifications, manage category codes, and monitor SKU distributions.

Summary:

- TOTAL CATEGORIES: 4
- ACTIVE CATEGORIES: 4
- TOTAL ASSIGNED SKU: 101 SKU
- UNASSIGNED / EMPTY: 0

Category Form:

CATEGORY NAME:

CATEGORY CODE:

STATUS:

SORT:

Category List:

CODE	CATEGORY NAME & DESCRIPTION	ASSIGNED SKU	STATUS	LAST UPDATED
CAT-ELEC-01	Electronics Electronic devices and components	24 SKU	ACTIVE	05/03/2024 23:46
CAT-OFFC-01	Office Supplies Stationery and general office consumables	28 SKU	ACTIVE	05/03/2024 23:46
CAT-RAW-01	Raw Materials Bulk materials used in manufacturing	25 SKU	ACTIVE	05/03/2024 23:46

Categories

Add New Category

CATEGORY CODE:

CATEGORY NAME:

DESCRIPTION:

STATUS:

Add New Category (dialog)

Warehouses

3 warehouses registered

Warehouse Form:

WAREHOUSE CODE:

WAREHOUSE NAME:

LOCATION:

STATUS:

Warehouse List:

CODE	NAME	LOCATION	STATUS
WH-BDG	Bandung Warehouse	Bandung, Indonesia	ACTIVE
WH-JKT	Jakarta Warehouse	Jakarta, Indonesia	ACTIVE
WH-SBY	Surabaya Warehouse	Surabaya, Indonesia	ACTIVE

Warehouses

Bandung Warehouse

CODE: WH-BDG | LOCATION: Bandung, Indonesia

STATUS: ACTIVE | CREATED AT: 2024-10-05 23:46

Stock Summary:

PRODUCTS STOCKED HERE: 100 | TOTAL STOCK QUANTITY: 3,400 Unit

Stock by Product:

SKU	PRODUCT NAME	CATEGORY	UNIT	STOCK	REORDER POINT
OFC-001	A4 Paper Ream	Office Supplies	ream	238 UNIT	100 Unit
RAW-001	Acrylic Sheet 3mm	Raw Materials	sheet	28 UNIT	20 Unit
SPR-006	Air Filter Regulator	Spares Parts	pcs	48 UNIT	20 Unit
RAW-002	Aluminum Rod 3m	Raw Materials	pcs	38 UNIT	20 Unit
RAW-008	Aluminum Sheet 2mm	Raw Materials	sheet	28 UNIT	12 Unit

Detail gudang: stok per produk

Suppliers

10 suppliers registered

Supplier Form:

SUPPLIER NAME:

CONTACT PERSON:

EMAIL:

STATUS:

Supplier List:

NAME	CONTACT PERSON	PHONE	EMAIL	STATUS
CV Chemika Solusi	Nurul Hidayah	022-5557990	info@chemika.solusi.co.id	ACTIVE
CV Mera Kertas Abadi	Siti Rahma	021-5553455	info@merakertas.co.id	ACTIVE
PT Raja Nusantara	Budi Santoso	022-5553456	purchasing@rajanusantara.co.id	ACTIVE
PT Raja Rangan Indonesia	Rudi Hermawan	027-5550123	info@bjarangan.id	ACTIVE
PT Logam Jaya	Ahmad Fauzi	024-5556789	sales@logampjaya.co.id	ACTIVE
PT Spangpar Prima	Hendra Gunawan	021-5556678	cs@spangparprima.co.id	ACTIVE

Suppliers

Customers

10 customers registered

Customer Form:

NAME:

CONTACT PERSON:

PHONE:

EMAIL:

STATUS:

Customer List:

NAME	CONTACT PERSON	PHONE	EMAIL	STATUS
CV Elektronik Favorit	Joko Susilo	024-6670123	order@elektronikfavorit.id	ACTIVE
CV Gudang Kita	Taufik Ismail	022-6666789	sales@gudangkita.id	ACTIVE
CV Toko Elektronik Modern	Fajar Nugroho	022-6663245	info@tokoelektronikmodern.co.id	ACTIVE
PT Distribusi Nasional	Yusuf Hidayat	031-6663456	order@distribusinasional.co.id	ACTIVE
PT Industri Manufaktur Sejahtera	Agus Setiawan	021-6666678	procurement@manufaktursejahtera.co.id	ACTIVE

Customers

► TAMPILKAN / LAKUKAN

- Buka menu **Master Data** satu per satu. Di tiap modul tunjukkan: daftar + pencarian/filter + paging, lalu **Add** → isi form → Simpan.
- Field form: **Categories** — kode, nama, deskripsi · **Warehouses** — kode, nama, lokasi · **Suppliers** — nama, kontak, email · **Customers** — nama, kontak, email, telepon.
- Edit** satu baris → ubah → Simpan. **Deactivate** → baris berstatus nonaktif dan tidak muncul di dropdown transaksi baru; **Activate** mengembalikannya.
- Categories**: tunjukkan **Export CSV** (mengikuti filter yang sedang dipakai) dan hapus sungguhan — kategori yang masih dipakai produk ditolak dihapus.
- Warehouses**: buka detail satu gudang → jumlah produk & total stok di gudang itu, tabel stok per produk (ada paging).

KATAKAN (baca naskah ini)

"Supplier dipakai Purchase Order, customer dipakai Sales Order, gudang menentukan di mana stok disimpan, dan kategori mengelompokkan produk. Semuanya bisa ditambah, diubah, dan dinonaktifkan oleh Admin saja."

Selip alasan — Hanya Admin yang mengelola — ucapkan di momen ini

"Tombol tambah dan ubah hanya muncul untuk Admin, tetapi yang menjaga sebenarnya server: setiap aksi tulis diperiksa izinnya dan token CSRF-nya. Role lain yang memaksa mengirim permintaan akan ditolak 403."

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTERS DATA

- Products
- Categories
- Warehouses
- Suppliers
- Customers

PROCUREMENT

- Purchase Orders

SALES

- Sales Orders

INVENTORY

- Stock Ledger

REPORTS

- Reports

ADMINISTRATION

- Users**

Users

5 users registered

+ Add User

NAME: Please Enter Name...

EMAIL: Please Enter Email Address...

ROLE: Select role...

STATUS: Select status...

Search Reset

NAME	EMAIL	ROLE	STATUS
Beri	sales1@example.com	Sales	ACTIVE
Grace	sales2@example.com	Sales	ACTIVE
Rita	admin@example.com	Admin	ACTIVE
Sari Demo	sari.demo@example.com	Sales	INACTIVE
Wawan	warehouse@example.com	Warehouse Staff	ACTIVE

Daftar Users

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTERS DATA

- Products
- Categories
- Warehouses
- Suppliers
- Customers

PROCUREMENT

- Purchase Orders

SALES

- Sales Orders

INVENTORY

- Stock Ledger

REPORTS

- Reports

ADMINISTRATION

- Users**

New User

NAME: Sari Demo

EMAIL: sari.demo@example.com

ROLE: Sales

PASSWORD: [password field]

Save Cancel

Form New User

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTERS DATA

- Products
- Categories
- Warehouses
- Suppliers
- Customers

PROCUREMENT

- Purchase Orders

SALES

- Sales Orders

INVENTORY

- Stock Ledger

REPORTS

- Reports

ADMINISTRATION

- Users**

New User

This email is already in use.

NAME: Sari Doka

EMAIL: sari.demo@example.com

ROLE: Sales

PASSWORD: [password field]

Save Cancel

Email kembar ditolak

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTERS DATA

- Products
- Categories
- Warehouses
- Suppliers
- Customers

PROCUREMENT

- Purchase Orders

SALES

- Sales Orders

INVENTORY

- Stock Ledger

REPORTS

- Reports

ADMINISTRATION

- Users**

Users

5 users registered

+ Add User

NAME: Please Enter Name...

EMAIL: Please Enter Email Address...

ROLE: Select role...

STATUS: Select status...

Search Reset

NAME	EMAIL	ROLE	STATUS
Beri	sales1@example.com	Sales	ACTIVE
Grace	sales2@example.com	Sales	ACTIVE
Rita	admin@example.com	Admin	ACTIVE
Sari Demo	sari.demo@example.com	Sales	INACTIVE
Wawan	warehouse@example.com	Warehouse Staff	ACTIVE

User dinonaktifkan (INACTIVE)

► TAMPILKAN / LAKUKAN

- Menu **Users** → daftar user dengan filter nama/email.
- **Add User**: nama, email (unik — huruf besar/kecil dianggap sama), role (Admin / Sales / WarehouseStaff), password awal (min. 6 karakter) → Simpan. Email kembar → ditolak.
- **Edit** user → ganti nama/role; kolom password dikosongkan berarti password lama tidak berubah.
- **Deactivate** user → ia tidak bisa login lagi (dan sesi yang sedang aktif ditolak pada request berikutnya); **Activate** memulihkan. Riwayat transaksinya tetap utuh.
- **Tunjukkan pengaman**: coba nonaktifkan atau ubah role akun Admin sendiri → ditolak ("You cannot deactivate your own account" / "You cannot change your own role").

🔪 KATAKAN (baca naskah ini)

"Di sinilah Admin mengatur siapa boleh masuk dan perannya. Pengguna tidak dihapus, hanya dinonaktifkan, supaya jejak transaksinya tetap ada. Admin juga tidak bisa mengunci dirinya sendiri."

💡 Selip alasan — Password & role — ucapkan di momen ini

"Password yang diketik Admin langsung di-hash bcrypt sebelum disimpan, jadi Admin pun tidak bisa melihat password siapa pun. Role hanya tiga pilihan tetap; tidak ada role buatan sendiri."

Langkah 3e — Purchase Order sisi Admin (buat, submit, batalkan)

Form Create PO

PO Draft — tombol Submit Order

Setelah Submit: Ordered (Receive Goods & Cancel Order)

PO dibatalkan (Cancelled)

► TAMPILKAN / LAKUKAN

- Menu **Purchase Orders** → daftar + filter status/nomor/supplier/gudang.
- **Create PO**: pilih supplier, gudang tujuan, tanggal, catatan; tambah baris produk (qty ≥ 1, harga beli ≥ 0) → Simpan → status **Draft**.
- Buka detail PO Draft → klik **Submit Order** → status **Ordered** (stepper Draft → Ordered → PartiallyReceived → Received ikut bergerak). Hanya Admin yang melihat tombol ini.
- Tunjuk **Cancel Order** pada PO yang belum menerima barang → konfirmasi → status Cancelled. PO yang sudah menerima barang tidak bisa dibatalkan.
- Setelah Warehouse menerima barang, kembali ke detail PO → tunjuk progres penerimaan (diterima vs dipesan) dan baris Stock Ledger-nya.

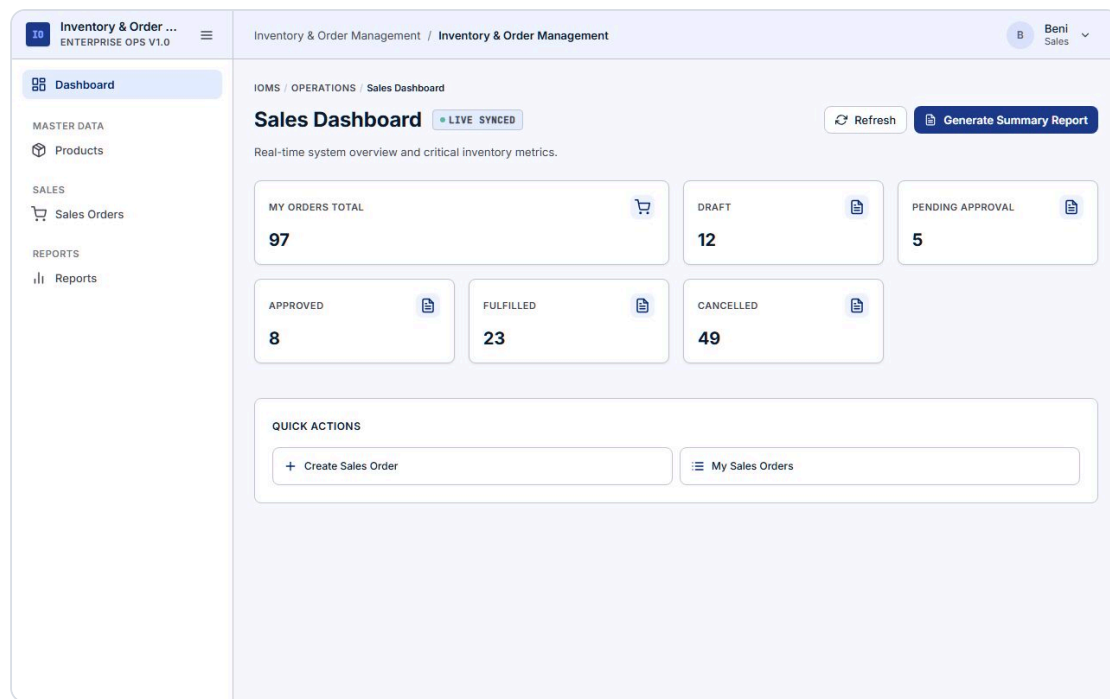
🔊 KATAKAN (baca naskah ini)

"Pembelian dimulai dari Purchase Order. Draft bisa dibuat Admin atau Warehouse, tetapi hanya Admin yang berhak mengirimnya ke supplier dan membatalkannya. Setelah barang datang, progres penerimaan terlihat di sini."



Selip alasan — Pemisahan wewenang di pembelian — ucapkan di momen ini

"Sama seperti penjualan, pembelian juga dipisah; pengusul dan pengesah tidak harus orang yang sama. Submit dan Cancel dijaga izin sendiri-sendiri yang hanya dimiliki Admin."



Tampilan aplikasi yang harus Anda tampilkan

► TAMPILKAN / LAKUKAN

- Login sebagai Sales (Beni). Tunjuk sidebar: hanya **Dashboard, Products (lihat saja), Sales Orders, Reports**.
- Dashboard Sales hanya menampilkan pesanan miliknya.

🎤 KATAKAN (baca naskah ini)

"Sekarang saya pindah peran menjadi Sales. Lihat bedanya — menunya jauh lebih sedikit. Sales fokus pada Sales Order, dan dashboardnya hanya menampilkan pesanan milik Sales ini sendiri."

💡 Selip alasan — Role journey — ucapkan di momen ini

"Inilah pembagian tugas: Sales fokus membuat pesanan. Data pesanan difilter berdasarkan ID user, jadi Sales tidak melihat pesanan orang lain."

Inventory & Order Management / Inventory & Order Management

Beni (upd) (upd) Sales

Dashboard

MASTER DATA

Products

SALES

Sales Orders

REPORTS

Reports

Back New Sales Order

ORDER DATE

04/10/2026

CUSTOMER

— Select —

SOURCE WAREHOUSE

— Select —

NOTE

Line Items

NAME	SKU	UNIT	QTY	SALE PRICE
—			1	0.00

Add Line

Cancel Save

Tampilan aplikasi yang harus Anda tampilkan

► TAMPILKAN / LAKUKAN

- Menu Sales Orders → tombol buat SO baru (atau Quick Action "Create Sales Order"). Pilih customer, gudang sumber, tanggal.
- Tambah baris barang: pilih produk, isi jumlah & harga. Simpan → jadi Draft.

🔊 KATAKAN (baca naskah ini)

"Tugas utama Sales: membuat pesanan. Saya pilih customer, gudang asal barang, lalu tambah barang beserta jumlah dan harga, lalu Simpan. Pesanan tersimpan sebagai Draft."

💡 Selip alasan — Transaksi — ucapkan di momen ini

"Saat saya tekan Simpan, data dikirim ke server, dicek dulu aturannya — customer dan gudang harus aktif, produk aktif, jumlah minimal 1. Lalu header pesanan dan semua barisnya disimpan dalam satu transaksi, supaya tidak ada pesanan dengan barang separuh. Perhatikan: stok belum dikurangi di sini — stok baru berubah nanti saat barang benar-benar dikeluarkan."

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

Products

SALES

Sales Orders

REPORTS

Reports

25 orders total

ORDER NUMBER:

CUSTOMER NAME:

STATUS:

WAREHOUSE:

Search Reset

ID	Date	CUSTOMER	WAREHOUSE	STATUS	ITEMS & QTY	TOTAL VALUE	View
#15	2020-10-05	UD Sumber Regis	Jakarta Warehouse	PENDING APPROVAL	2 items 20 units	Rp 1.220.000	View
#11	2020-10-05	PT Industri Manufaktur Segal	Jakarta Warehouse	PENDING APPROVAL	1 item 2 units	Rp 90.000	View
#7	2020-10-05	PT Industri Manufaktur Segal	Bandung Warehouse	DRAFT	1 item 12 units	Rp 1.068.000	View
#5	2020-10-05	PT Karya Cerdas Indonesia	Jakarta Warehouse	DRAFT	2 items 20 units	Rp 1.480.000	View
#3	2020-10-05	PT Retail Maju Bersama	Jakarta Warehouse	DRAFT	2 items 10 units	Rp 917.000	View
#1	2020-10-05	CV Teknik Elektronik Modern	Jakarta Warehouse	DRAFT	1 item	Rp 175.000	View

Daftar SO milik sendiri

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

Products

SALES

Sales Orders

REPORTS

Reports

SO-0051 DRAFT

CV Elektronik Favorit Order Date: 2020-10-05 Created by Beni

1 Draft 2 Pending Approval 3 Approved 4 Fulfilled

Sales Order & Stock Allocation Protocol: Approved Sales Orders place a hard stock reservation against the assigned Source Warehouse. Physical inventory is deducted from Product Stock only upon validated Goods Issue, recording an immutative audit entry in the Stock Ledger.

CUSTOMER: CV Elektronik Favorit Customer master record verified

SOURCE FACILITY: Bandung Warehouse Physical inventory deducted here upon Goods Issue.

AUDIT & CHAIN OF CUSTODY Order Date: 2020-10-05 Stock Ledger Entries: 0

Line Items 1 LINE

PRODUCT	SKU	QTY	UNIT PRICE	LINE SUBTOTAL
A4 Paper Ream	CFC-001	3	Rp 50.000	Rp 150.000
Distinct SKUs: 1 - Total Quantity: 3 units				Rp 150.000

Edit Draft Submit for Approval Cancel Order Back

Detail Draft: Edit Draft / Submit / Cancel

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

Products

SALES

Sales Orders

REPORTS

Reports

Edit Draft Sales Order

ORDER DATE: 10/05/2020

CUSTOMER: CV Elektronik Favorit

SOURCE WAREHOUSE: Bandung Warehouse

NOTE

Line Items

NAME	SKU	UNIT	QTY	SALE PRICE
A4 Paper Ream	CFC-001	INAH	5	50000.00

Add Line Cancel Save

Edit Draft

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

Products

SALES

Sales Orders

REPORTS

Reports

New Sales Order

ORDER DATE: 10/05/2020

CUSTOMER: CV Elektronik Favorit

SOURCE WAREHOUSE: Bandung Warehouse

NOTE

Line Items

NAME	SKU	UNIT	QTY	SALE PRICE
A4 Paper Ream	CFC-001	INAH	0	50000

Add Line Cancel Save

Validasi (qty 0 ditolak)

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

Products

SALES

Sales Orders

REPORTS

Reports

SO-0023 DRAFT

CV Elektronik Favorit Order Date: 2020-10-05 Created by Beni

1 Draft 2 Pending Approval 3 Approved 4 Fulfilled

Sales Order & Stock Allocation Protocol: Approved Sales Orders place a hard stock reservation against the assigned Source Warehouse. Physical inventory is deducted from Product Stock only upon validated Goods Issue, recording an immutative audit entry in the Stock Ledger.

CUSTOMER: CV Elektronik Favorit Customer master record verified

SOURCE FACILITY: Jakarta Warehouse Physical inventory deducted here upon Goods Issue.

AUDIT & CHAIN OF CUSTODY Order Date: 2020-10-05 Stock Ledger Entries: 0

Line Items 1 LINE

PRODUCT	SKU	QTY	UNIT PRICE	LINE SUBTOTAL
A4 Paper Ream	CFC-001	2	Rp 50.000	Rp 100.000
Distinct SKUs: 1 - Total Quantity: 2 units				Rp 100.000

Edit Draft Submit for Approval Cancel Order Back

Cancel this sales order? This cannot be undone.

Keep order Yes, cancel order

Konfirmasi Cancel Order

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

Products

SALES

Sales Orders

REPORTS

Reports

SO-0023 APPROVED

PT Distribusi Nasional Order Date: 2020-10-04 Created by Beni

1 Draft 2 Pending Approval 3 Approved 4 Fulfilled

Sales Order & Stock Allocation Protocol: Approved Sales Orders place a hard stock reservation against the assigned Source Warehouse. Physical inventory is deducted from Product Stock only upon validated Goods Issue, recording an immutative audit entry in the Stock Ledger.

CUSTOMER: PT Distribusi Nasional Send SO 23 Approved

SOURCE FACILITY: Jakarta Warehouse Physical inventory deducted here upon Goods Issue.

AUDIT & CHAIN OF CUSTODY Order Date: 2020-10-04 Approved By: Rita (2020-10-05 00:00) Stock Ledger Entries: 0

Line Items 1 LINE

PRODUCT	SKU	QTY	UNIT PRICE	LINE SUBTOTAL
Roller Coaster Sheet	RAN-006	2	Rp 65.000	Rp 130.000
Distinct SKUs: 1 - Total Quantity: 2 units				Rp 130.000

Back

SO Approved: tidak ada tombol Cancel

► TAMPILKAN / LAKUKAN

- Menu **Sales Orders** → daftar hanya berisi SO milik Sales ini. Tunjuk filter status, nomor SO, nama customer, gudang, dan urutan tanggal; tombol Search / Reset; paging.
- Buka SO berstatus **Draft** → klik **Edit Draft** → ubah qty, tambah/hapus baris, atau ganti customer → Simpan. (Edit hanya mungkin saat Draft.)
- **Tunjukkan validasi:** simpan tanpa satu pun baris item, atau qty 0 → ditolak dengan pesan.
- Buka SO Draft/Pending lain → klik **Cancel Order** → dialog konfirmasi → status **Cancelled**.
- Buka SO yang sudah **Approved** → tombol Cancel tidak ada (pembatalan setelah disetujui harus lewat Admin).

"Sebelum diajukan, pesanan masih bisa saya perbaiki. Kalau salah input, saya edit saja. Kalau pelanggan batal sebelum disetujui, saya bisa membatalkannya sendiri. Tetapi begitu sudah disetujui, hanya Admin yang dapat membatalkan."

 Selip alasan — Kepemilikan data — ucapkan di momen ini

"Saya hanya bisa mengubah dan membatalkan pesanan milik saya sendiri. Kalau saya mencoba membuka pesanan Sales lain lewat alamat URL, server menjawab 403."

Inventory & Order Management / Inventory & Order Management

IOMS / SALES ORDERS / SO-0508

SO-0508 DRAFT

CV Elektronik Favorit - Order Date: 2026-10-04 - Created by Beni (upd) (upd)

1 Draft — 2 Pending Approval — 3 Approved — 4 Fulfilled

Sales Order & Stock Allocation Protocol. Approved Sales Orders place a hard stock reservation against the assigned Source Warehouse. Physical inventory is deducted from ProductStock only upon validated Goods Issue, recording an immutable audit entry in the Stock Ledger.

CUSTOMER
CV Elektronik Favorit
Customer master record verified

SOURCE FACILITY
Bandung Warehouse
Physical inventory deducted here upon Goods Issue.

AUDIT & CHAIN OF CUSTODY
Order Date: 2026-10-04
Stock Ledger Entries: 0

Line Items 1 LINE

PRODUCT	SKU	QTY	UNIT PRICE	LINE SUBTOTAL
A4 Paper Ream	0FC-001 (EDITED) (EDITED) (EDI	1	Rp 55,000	Rp 55,000
Distinct SKUs: 1 - Total Quantity: 1 unit				Rp 55,000

Submit for Approval Cancel Order Back

Tampilan aplikasi yang harus Anda tampilkan

► TAMPILKAN / LAKUKAN

- Buka detail SO yang Draft. Tunjuk tombol **Submit for Approval** (juga ada **Edit Draft** dan **Cancel Order**).
- Klik Submit → status berubah ke "Pending Approval".

🎤 KATAKAN (baca naskah ini)

"Setelah pesanan siap, saya klik Submit. Statusnya berubah menjadi Menunggu Persetujuan. Sekarang pesanan ini keluar dari tangan Sales dan menunggu Admin — ini perpindahan pertama antar-role."

💡 Selip alasan — Handoff — ucapkan di momen ini

"Sales tidak bisa menyetujui pesannya sendiri. Jadi di titik ini tugas berpindah ke Admin."

Inventory & Order Management / Inventory & Order Management

Beni (upd) (upd) Sales

Dashboard

MASTER DATA

Products

SALES

Sales Orders

REPORTS

Reports

Back New Sales Order

ORDER DATE

04/10/2026

CUSTOMER

— Select —

SOURCE WAREHOUSE

— Select —

NOTE

Line Items

NAME	SKU	UNIT	QTY	SALE PRICE
—			1	0.00

Add Line

Cancel Save

Tampilan aplikasi yang harus Anda tampilkan

► TAMPILKAN / LAKUKAN

- Masih sebagai Sales, buka pesanan yang Pending. Tunjuk: TIDAK ada tombol Approve/Reject (hanya Cancel Order).

🎤 KATAKAN (baca naskah ini)

"Perhatikan baik-baik: sebagai Sales, di pesanan yang menunggu persetujuan ini, saya tidak punya tombol Approve. Ini sengaja."

💡 Selip alasan — Segregation of Duties — ucapkan di momen ini

"Ini adalah Segregation of Duties — Sales tidak boleh menyetujui pesanan apa pun, termasuk miliknya sendiri. Ini aturan anti-kecurangan, dan ditegakkan di server lewat tabel izin `role_permissions` dan `SalesOrderPolicy`, bukan sekadar menyembunyikan tombol."

Export Preview

TYPE	ORDER NO.	DATE	CUSTOMER/SUPPLIER	WAREHOUSE	STATUS	ITEMS COUNT	TOTAL VALUE	CREATED BY
SO	3	2026-10-05	PT Retail Maju Bersama	Jakarta Warehouse	Draft	2	\$17000.00	Beni
SO	52	2026-10-05	CV Elektronik Farant	Bandung Warehouse	Cancelled	1	100000.00	Beni
SO	51	2026-10-05	CV Elektronik Farant	Bandung Warehouse	Pending Approval	1	250000.00	Beni
SO	15	2026-10-05	UD Sumber Rajek	Jakarta Warehouse	Pending Approval	2	1220000.00	Beni
SO	11	2026-10-05	PT Industri Manufaktur Segit	Jakarta Warehouse	Pending Approval	1	90000.00	Beni
SO	7	2026-10-05	PT Industri Manufaktur Segit	Bandung Warehouse	Draft	1	1068000.00	Beni
SO	1	2026-10-05	CV Toko Elektronik Modern	Jakarta Warehouse	Draft	1	375000.00	Beni
SO	5	2026-10-05	PT Karya Cendek Indonesia	Jakarta Warehouse	Draft	2	1480000.00	Beni
SO	9	2026-10-04	PT Distribusi Nasional	Jakarta Warehouse	Pending Approval	1	1377000.00	Beni
SO	23	2026-10-04	PT Distribusi Nasional	Jakarta Warehouse	Approved	1	130000.00	Beni

Reports Sales: preview + Export Orders (milik sendiri)

Sales Dashboard

MY ORDERS TOTAL: 25

DRAFT: 4

PENDING APPROVAL: 6

APPROVED: 6

FULFILLED: 5

CANCELLED: 4

QUICK ACTIONS

+ Create Sales Order

My Sales Orders

Dashboard Sales

Products

Manage catalog items, category classifications, and master inventory units.

TOTAL PRODUCTS (SKU): 100 SKU

NORMAL STOCK ITEMS: 99 SKU

LOW STOCK WARNING: 1 SKU

OUT OF STOCK: 0 SKU

Sort Products: Name A-Z

Product Name: Please Enter Product Name...

Category: Select category...

Warehouse: Select warehouse...

Search

SKU CODE	PRODUCT NAME	UOM	PRICE (RP)	TOTAL PHYSICAL STOCK	STATUS
OTC - 001	A4 Paper 80mm Office Supplies	sheet	Rp 50.000	280 sheet	IN STOCK
...	...	...	...	...	...

Products (read-only)

My Profile

View and update your account information.

Beni Sales • Joined Oct 2026

Account Information

Full Name: Beni

Email Address: sales1@example.com

Role: Sales

Cancel / Back

Save Changes

My Profile

► TAMPILKAN / LAKUKAN

- Menu **Reports** → halaman versi ringkas: pilih tanggal mulai & akhir (wajib, mulai $\leq$ akhir) → **Preview** (tabel contoh isi file) → **Export Orders** → CSV.
- Tunjuk bahwa isi CSV hanya **pesanan milik Sales ini**; tidak ada opsi Stock Ledger maupun Purchase Order.
- Dashboard Sales: kartu **My Orders Total** dan jumlah per status (Draft, Pending Approval, Approved, Fulfilled, Cancelled); Quick Actions "Create Sales Order" dan "My Sales Orders".
- Menu **Products**: cari/filter katalog dan lihat harga jual & stok — tanpa tombol tambah/ubah.
- Menu akun → **My Profile** → ubah nama/email → Simpan → **Logout**.

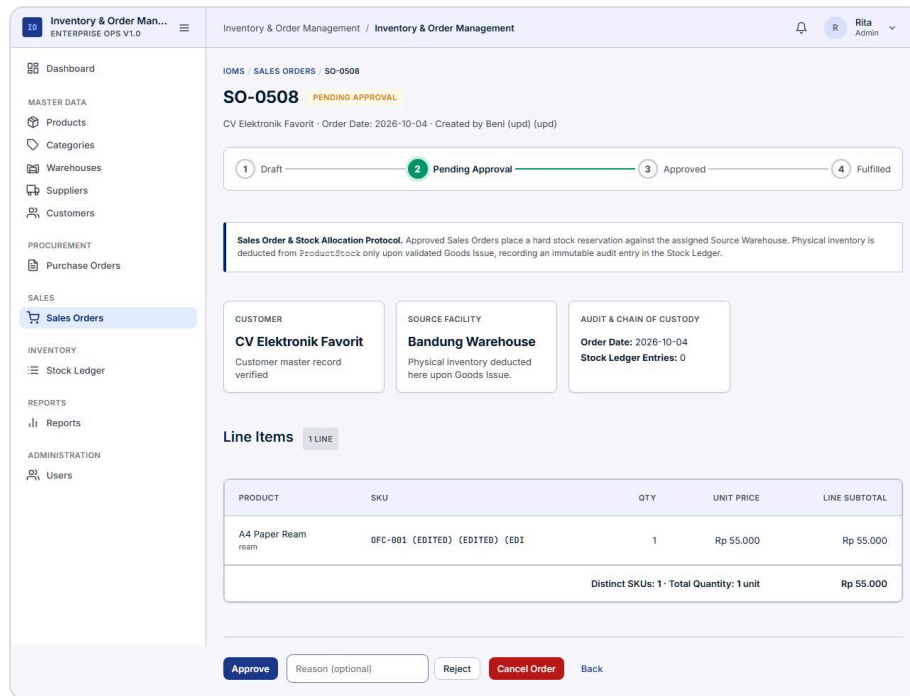
🔊 KATAKAN (baca naskah ini)

"Sales juga bisa mengambil laporan pesannya sendiri dalam bentuk CSV untuk rekap. Yang diekspor hanya pesanan saya — bukan seluruh perusahaan."



Selip alasan — Data dibatasi per pemilik — ucapkan di momen ini

"Pembatasan ini ada di server: saat mengekspor, query difilter berdasarkan ID user yang login. Bukan sekadar menyembunyikan pilihan di layar."



Tampilan aplikasi yang harus Anda tampilkan

► TAMPILKAN / LAKUKAN

- Login sebagai Admin. Buka pesanan yang sama (Pending Approval).
- Sekarang MUNCUL tombol Approve / Reject. Klik Approve → status jadi Approved.

🗣 KATAKAN (baca naskah ini)

"Saya kembali menjadi Admin, dan membuka pesanan yang tadi. Lihat — sekarang tombol Approve dan Reject muncul, karena hanya Admin yang berhak. Saya klik Approve, dan statusnya menjadi Disetujui. Dicatat juga siapa yang menyetujui dan kapan, untuk audit."

💡 Selip alasan — Otorisasi server — ucapkan di momen ini

"Jadi role berbeda, pesanan yang sama persis — tapi tombolnya berbeda tergantung siapa yang login. Itu bukti otorisasi bekerja di level server. Sekarang pesanan siap diserahkan ke Warehouse untuk dikeluarkan barangnya." (Catatan: Reject mengubah status menjadi Cancelled dengan alasan opsional — tidak ada status "Rejected" terpisah.)

Langkah 7b — Reject, Cancel, dan kelola Sales Order sebagai Admin

Sales Orders

50 orders total

ORDER NUMBER: CUSTOMER NAME:

STATUS: WAREHOUSE:

ID	Date	CUSTOMER	WAREHOUSE	STATUS	ITEMS / QTY	TOTAL VALUE	
#20	2026-10-05	PT Retail Maju Bersama	Bandung Warehouse	PENDING APPROVAL	1 line 20 units	Rp 1.050.000	View
#15	2026-10-05	UD Sumber Rejeki	Jakarta Warehouse	PENDING APPROVAL	2 line 20 units	Rp 1.220.000	View
#11	2026-10-05	PT Industri Manufaktur Segel	Jakarta Warehouse	PENDING APPROVAL	1 line 20 units	Rp 90.000	View
#8	2026-10-05	CV Gudang Kita	Jakarta Warehouse	DRAFT	1 line 5 units	Rp 525.000	View
#7	2026-10-05	PT Industri Manufaktur Segel	Bandung Warehouse	DRAFT	1 line 12 units	Rp 1.050.000	View
#6	2026-10-05	PT Industri Manufaktur Segel	Nekarya Warehouse	DRAFT	1 line	Rp 5.000.000	View

Daftar SO semua Sales

SO-0020 PENDING APPROVAL

PT Retail Maju Bersama - Order Date: 2026-10-05 - Created by Grace

1 Draft 2 Pending Approval 3 Approved 4 Fulfilled

Sales Order & Stock Allocation Protocol: Approved Sales Orders place a hard stock reservation against the assigned Source Warehouse. Physical inventory is deducted from Product Stock only upon validated Goods Issue, recording an immutuable audit entry in the Stock Ledger.

CUSTOMER: PT Retail Maju Bersama
Seed SO ID: 0020

SOURCE FACILITY: Bandung Warehouse
Physical inventory deducted here upon Goods Issue.

AUDIT & CHAIN OF CUSTODY:
Order Date: 2026-10-05
Stock Ledger Entries: 0

Line Items 1 LINE

PRODUCT	SKU	QTY	UNIT PRICE	LINE SUBTOTAL
Paper Cutter A4	GFC-817	25	Rp 42.000	Rp 1.050.000
Distinct SKUs: 1 - Total Quantity: 25 units				Rp 1.050.000

Detail SO Pending: kolom alasan + Reject

SO-0020 CANCELLED

PT Retail Maju Bersama - Order Date: 2026-10-05 - Created by Grace

1 Draft 2 Pending Approval 3 Approved 4 Fulfilled 5 Cancelled

Sales Order & Stock Allocation Protocol: Approved Sales Orders place a hard stock reservation against the assigned Source Warehouse. Physical inventory is deducted from Product Stock only upon validated Goods Issue, recording an immutuable audit entry in the Stock Ledger.

CUSTOMER: PT Retail Maju Bersama
Seed SO ID: 0020

SOURCE FACILITY: Bandung Warehouse
Physical inventory deducted here upon Goods Issue.

AUDIT & CHAIN OF CUSTODY:
Order Date: 2026-10-05
Stock Ledger Entries: 0

Line Items 1 LINE

PRODUCT	SKU	QTY	UNIT PRICE	LINE SUBTOTAL
Paper Cutter A4	GFC-817	25	Rp 42.000	Rp 1.050.000
Distinct SKUs: 1 - Total Quantity: 25 units				Rp 1.050.000

Setelah Reject: Cancelled

SO-0008 DRAFT

CV Gudang Kita - Order Date: 2026-10-05 - Created by Grace

1 Draft 2 Pending Approval 3 Approved 4 Fulfilled

Sales Order & Stock Allocation Protocol: Approved Sales Orders place a hard stock reservation against the assigned Source Warehouse. Physical inventory is deducted from Product Stock only upon validated Goods Issue, recording an immutuable audit entry in the Stock Ledger.

CUSTOMER: CV Gudang Kita
Seed SO ID: 0008

SOURCE FACILITY: Jakarta Warehouse
Physical inventory deducted here upon Goods Issue.

AUDIT & CHAIN OF CUSTODY:
Order Date: 2026-10-05
Stock Ledger Entries: 0

Line Items 1 LINE

PRODUCT	SKU	QTY	UNIT PRICE	LINE SUBTOTAL
Gaming Mousepad XL	ELEK-819	5	Rp 105.000	Rp 525.000
Distinct SKUs: 1 - Total Quantity: 5 units				Rp 525.000

Admin melihat SO Draft milik Sales

► TAMPILKAN / LAKUKAN

- Menu **Sales Orders** → Admin melihat **semua** SO dari semua Sales. Filter status *Pending Approval* untuk antrean persetujuan; juga filter nomor SO, customer, gudang, urutan.
- Buka SO Pending lain → klik **Reject**, isi alasan (opsional tapi disarankan) → status menjadi **Cancelled** dengan alasan tersimpan (tidak ada status "Rejected" terpisah; Sales perlu membuat SO baru).
- Admin bisa **Edit Draft** milik Sales mana pun dan **Cancel Order** pada SO yang belum Fulfilled. SO Fulfilled tidak bisa dibatalkan oleh siapa pun.
- Admin juga boleh membuat SO sendiri dan menyetujuinya — larangan self-approval berlaku untuk role Sales, bukan Admin.

🎤 KATAKAN (baca naskah ini)

"Sebagai Admin saya punya keputusan akhir: menyetujui, menolak dengan alasan, atau membatalkan. Semua keputusan tercatat — siapa dan kapan."

💡 **Selip alasan — Aturan status — ucapkan di momen ini**

"Setiap perubahan status mengikuti aturan yang tetap: hanya Pending yang bisa disetujui/ditolak, hanya Approved yang bisa dikeluarkan barangnya, dan yang sudah Fulfilled tidak bisa dibatalkan — siapa pun yang mencoba, termasuk Admin."

SKU	PRODUCT NAME	WAREHOUSE	TYPE	QTY	STOCK BEFORE	AFTER
RAM-011	Carbon Steel Bar 20mm	Jakarta Warehouse	RECEIPT	+4	62	66
RAM-018	HDPE Sheet 5mm	Jakarta Warehouse	ISSUE	-10	65	55
OFC-007	Correction Tape	Bandung Warehouse	ISSUE	-6	38	24
ELEC-008	LED Monitor 24in	Jakarta Warehouse	ISSUE	-2	58	48

Stock Ledger (filter + tabel)

Reports: parameter, analitik, tabel

My Profile

Menu akun: My Profile & Logout

► TAMPILKAN / LAKUKAN

- Menu **Stock Ledger**: filter SKU, nama produk, jenis (Receipt / Issue / Adjustment), gudang; urutkan kolom; 25 baris per halaman. Tunjuk baris Receipt dan Issue dari demo tadi.
- Menu **Reports**: pilih jenis laporan — *Stock Valuation & Turnover, Inventory Aging, Slow-Moving & Dead Stock, Movement Ledger Audit* — atur rentang tanggal, gudang, kategori, pencarian, urutan → **Generate**. Tunjuk grafik & tabel.
- Klik **Export Stock Ledger** → CSV. Isi tanggal mulai & akhir → **preview** lalu **Export Orders** → CSV gabungan SO (semua Sales) + semua PO. Tanggal kosong atau mulai > akhir → ditolak.
- Menu akun (pojok kanan atas) → **My Profile** → ubah nama/email → muncul pesan sukses. Lalu **Logout**.

🔪 KATAKAN (baca naskah ini)

"Untuk pelaporan, Admin punya analitik stok dan bisa mengekspor data ke CSV untuk dianalisis di luar. Kartu stok adalah rekaman mutlak semua pergerakan barang."

💡 **Selip alasan — Jejak audit & ekspor — ucapkan di momen ini**

"Ekspor memakai filter yang sama dengan layar, jadi yang diunduh sama dengan yang terlihat. Ganti password tidak lewat profil; Admin melakukannya lewat menu Users."

The screenshot displays the 'Warehouse Dashboard' interface. It includes a sidebar with navigation links: Dashboard, Products, Purchase Orders, Stock Ledger, and Reports. The main content area features a 'Warehouse Dashboard' header with a 'Refresh' button and a 'Generate Summary Report' button. Below this, there are two summary cards: 'GOODS RECEIPT QUEUE' with a value of 29 and 'GOODS ISSUE QUEUE' with a value of 15. A 'BELOW REORDER POINT' indicator shows a value of 1. The dashboard is divided into two main sections: 'Goods Issue Queue — Approved Sales Orders' and 'Goods Receipt Queue — Purchase Orders Awaiting Delivery'. Each section contains a table with columns for ID, DATE, CUSTOMER, SOURCE WAREHOUSE, LINES, and a status (e.g., 'Process Issue' or 'Receive'). The 'Goods Issue Queue' table lists 12 items, and the 'Goods Receipt Queue' table lists 12 items. At the bottom, there is a 'Low Stock Products' section with a table showing items like 'OPIC 007' with a 'Current Stock' of 74 and a 'Reorder Point' of 90.

Tampilan aplikasi yang harus Anda tampilkan

► TAMPILKAN / LAKUKAN

- Login sebagai Warehouse (Wawan). Sidebar: **Dashboard, Products (lihat saja), Purchase Orders, Stock Ledger, Reports**. Tidak ada menu Sales Orders.
- Pintu masuk ke pengeluaran barang adalah Dashboard: tabel **Goods Issue Queue** (SO Approved, tombol **Process Issue**) dan **Goods Receipt Queue** (PO menunggu barang, tombol **Receive**).

🎤 KATAKAN (baca naskah ini)

"Sekarang saya menjadi Warehouse Staff. Tugas saya menangani barang fisik: menerima barang dari pembelian, dan mengeluarkan barang untuk pesanan yang sudah disetujui. Antrean kerjanya langsung ada di dashboard."

💡 Selip alasan — Kartu Stok — ucapkan di momen ini

"Di sinilah stok benar-benar berubah. Setiap pergerakan barang dicatat di Kartu Stok — aplikasi tidak menyediakan fitur untuk mengedit atau menghapus barisnya — supaya selalu bisa diaudit."

Inventory & Order Management / Inventory & Order Management

PO-0173 **PARTIALLY RECEIVED**

Order Date: 2026-10-04 - Created by Rita - Destination: Bandung Warehouse

[Receive Goods](#) [Cancel Order](#) [Back](#)

1 Draft 2 Ordered 3 **Partially Received** 4 Received

Enterprise Stock Governance Protocol: Goods receipt increments product_stock_qty and writes a matching immutable stock_ledger row in one DB transaction. Stock quantities cannot be edited directly here.

Inbound Intake & Fulfillment Progress

Received: 8 pcs (80%) | Remaining: 2 pcs

TOTAL ORDERED	TOTAL RECEIVED	REMAINING TO RECEIVE
10	8 80% Verified at Dock	2

SUPPLIER
CV Chemika Solusi
Procurement faksim registered in master data

DESTINATION FACILITY
Bandung Warehouse
Inbound goods are recorded against this facility's product_stock_qty

AUDIT & SCHEDULE
Order Date: 2026-10-04
Created By: Rita
Stock Ledger Entries: 1

Line Items

PRODUCT	SKU	ORDERED QTY	RECEIVED QTY	REMAINING	UNIT PRICE	LINE SUBTOTAL
USB-C Cable 1m	ELEC-883	10 pcs	8	2	Rp 10,000	Rp 100,000
Total Purchase Value						Rp 100,000

Subtotal strictly governed by Unit Price × Ordered Qty. Tax & logistics calculated separately.

Audit Log

Every Stock Ledger entry referencing this purchase order. The ledger is append-only — entries cannot be edited or deleted.

Receipt +8 units of USB-C Cable 1m at Bandung Warehouse
2026-10-04 13:09 - by Wawan

Tampilan aplikasi yang harus Anda tampilan

► TAMPILKAN / LAKUKAN

- Buka Purchase Order berstatus Ordered (dari antrean dashboard → **Receive**, atau Purchase Orders → detail → **Receive Goods**). Isi jumlah diterima per baris, simpan.
- Tunjuk: stok produk NAIK, muncul baris di Kartu Stok bertipe Receipt.

🎤 KATAKAN (baca naskah ini)

"Pertama, menerima barang dari pemasok lewat Purchase Order. Saya isi jumlah yang diterima lalu simpan. Stok produk naik, dan muncul catatan baru di Kartu Stok."

💡 Selip alasan — Transaksi — ucapkan di momen ini

"Penambahan stok dan pencatatan Kartu Stok terjadi dalam satu transaksi — kalau salah satu gagal, dua-duanya dibatalkan, jadi stok dan catatannya tidak pernah beda."

Langkah 9b — Buat PO usulan, terima sebagian, dan produk

Products
Manage catalog items, category classifications, and master inventory units.

TOTAL PRODUCTS (SKU)
100 SKU
After across all warehouses

NORMAL STOCK ITEMS
99 SKU
Stock above minimum threshold

LOW STOCK WARNING
1 SKU
Approaching reorder point

OUT OF STOCK
0 SKU
Reorder required immediately

Sort Products
Name A-Z

Product SKU / BARCODE
Please Enter SKU Code or Barcode...

Product Name
Please Enter Product Name...

Category
Select category...

Stock Status
All Active

Warehouse
Select warehouse...

Search | **Reset**

SKU CODE	PRODUCT NAME	UOM	PRICE (RP)	TOTAL PHYSICAL STOCK	STATUS
SFC-001	A4 Paper Ream Office Supplies	Ream	Rp 15,000	280	In Stock
...	...	...	...	...	...

Products read-only

Master Data | **PURCHASE ORDERS** | **Create Purchase Order**

Order Details

SUPPLIER *
CV Chemika Solusi

DESTINATION WAREHOUSE *
Bandung Warehouse — Bandung, Indonesia

ORDER DATE *
10/05/2026

NOTE
Usulan restock dari Warehouse

Line Items

Create PO usulan

PO-0053 (DRAFT)
Order Date: 2026-10-05 | Created by: Wisnu | Destination: Bandung Warehouse

Progress
1 Draft — 2 Ordered — 3 Partially Received — 4 Received

Enterprise Stock Governance Protocol
Goods receipt increments product_stocks, quant1ty and writes a matching immutable stock_ledger row in one DB transaction. Stock quantities cannot be edited directly here.

Inbound Intake & Fulfillment Progress

Received: 0 item (0%) | Remaining: 20 item

TOTAL ORDERED	TOTAL RECEIVED	REMAINING TO RECEIVE
20	0	20

SUPPLIER
CV Chemika Solusi
Usulan restock dari Warehouse

DESTINATION FACILITY
Bandung Warehouse
Inbound goods are recorded against this facility's product_stocks.

AUDIT & SCHEDULE
Order Date: 2026-10-05
Created By: Wisnu
Stock Ledger Entries: 0

Line Items

PO Draft: tanpa Submit/Cancel

Receive Goods — #27

PRODUCT	QTY ORDERED	QTY RECEIVED	QTY REMAINING	RECEIVE NOW
RAW011 — Carbon Steel Bar 20mm	25 pcs	12	13	4

Enter the quantity received now for each line. This cannot exceed the remaining quantity, and updates stock immediately and permanently once confirmed.

Confirm Receipt | **Cancel**

Form Receive Goods (isi sebagian)

PO-0027 PARTIALLY RECEIVED
Order Date: 2026-09-28 | Created by: Risa | Destination: Jakarta Warehouse

Progress
1 Draft — 2 Ordered — 3 Partially Received — 4 Received

Enterprise Stock Governance Protocol
Goods receipt increments product_stocks, quant1ty and writes a matching immutable stock_ledger row in one DB transaction. Stock quantities cannot be edited directly here.

Inbound Intake & Fulfillment Progress

Received: 16 pcs (64%) | Remaining: 9 pcs

TOTAL ORDERED	TOTAL RECEIVED	REMAINING TO RECEIVE
25	16	9

SUPPLIER
PT Logam Jaya
Send PO 27 Partial

DESTINATION FACILITY
Jakarta Warehouse
Inbound goods are recorded against this facility's product_stocks.

AUDIT & SCHEDULE
Order Date: 2026-09-28
Created By: Risa
Stock Ledger Entries: 2

Line Items

PartiallyReceived + audit log

Warehouse Dashboard | **Warehouse Dashboard** | **Generate Summary Report**

GOODS RECEIPT QUEUE
21

GOODS ISSUE QUEUE
12

BELOW REORDER POINT
1

Goods Issue Queue — Approved Sales Orders

SO	DATE	CUSTOMER	SOURCE WAREHOUSE	LINE#	Process Issue
#26	2026-09-30	PT Industri Manufaktur Sejahtera	Jakarta Warehouse	1	Process Issue
#24	2026-10-01	PT NusantaraRetail	Bandung Warehouse	2	Process Issue
#29	2026-10-01	CV Toko Elektronik Modern	Bandung Warehouse	2	Process Issue
#21	2026-10-02	CV Toko Elektronik Modern	Jakarta Warehouse	1	Process Issue
#27	2026-10-02	CV Gudang Kita	Bandung Warehouse	1	Process Issue
#31	2026-10-02	UD Prima Bakti	Bandung Warehouse	1	Process Issue
#22	2026-10-03	UD Sumber Rezeki	Bandung Warehouse	1	Process Issue

Bel notifikasi stok rendah

TAMPILKAN / LAKUKAN

- Menu **Products** → lihat stok per gudang dan produk low stock (read-only, tanpa tombol Add/Edit).
- Menu **Purchase Orders** → **Create PO** (supplier, gudang tujuan, tanggal, baris produk & qty) → status **Draft**. Buka detailnya: tidak ada tombol Submit Order maupun Cancel Order — keduanya khusus Admin.
- Buka PO **Ordered** → **Receive Goods** → isi hanya *sebagian* qty → Simpan → status **PartiallyReceived**, stok naik sebesar qty itu, progres penerimaan di detail PO bertambah.
- Ulangi Receive untuk sisanya → status **Received**. Coba isi qty melebihi sisa → ditolak.
- Di dashboard, tunjuk **bel notifikasi** stok rendah dan tombol **Mark All as Read**.

KATAKAN (baca naskah ini)

"Warehouse bisa mengusulkan pembelian ketika stok menipis, tetapi pengirimannya ke supplier harus dilakukan Admin. Saat barang datang bertahap, saya terima sesuai jumlah fisik — sisanya tetap tercatat sebagai belum diterima."

Selip alasan — Penerimaan parsial — ucapkan di momen ini

"Setiap penerimaan menambah stok dan menulis baris Kartu Stok dalam satu transaksi. Jumlah yang diterima tidak boleh melebihi sisa pesanan, sehingga stok tidak bisa digelembungkan."

The screenshot shows a 'Stock Ledger' application window. It has a sidebar with navigation options like 'Dashboard', 'Inventory', 'Stock Ledger', 'Reports', 'Settings', and 'Users'. The main area displays a table of stock transactions. The table has columns for 'ID', 'PRODUCT NAME', 'QUANTITY', 'UNIT', 'DATE', 'AMOUNT', and 'STATUS'. The transactions are listed in a table with alternating light and dark blue rows. The status of each transaction is indicated by a colored dot: green for 'Selesai' (Completed) and red for 'Masuk' (Incoming).

ID	PRODUCT NAME	QUANTITY	UNIT	DATE	AMOUNT	STATUS
101	STOK BARANG	100	kg	2023-10-20	100000	Selesai
102	STOK BARANG	100	kg	2023-10-20	100000	Selesai
103	STOK BARANG	100	kg	2023-10-20	100000	Selesai
104	STOK BARANG	100	kg	2023-10-20	100000	Selesai
105	STOK BARANG	100	kg	2023-10-20	100000	Selesai
106	STOK BARANG	100	kg	2023-10-20	100000	Selesai
107	STOK BARANG	100	kg	2023-10-20	100000	Selesai
108	STOK BARANG	100	kg	2023-10-20	100000	Selesai
109	STOK BARANG	100	kg	2023-10-20	100000	Selesai
110	STOK BARANG	100	kg	2023-10-20	100000	Selesai
111	STOK BARANG	100	kg	2023-10-20	100000	Selesai
112	STOK BARANG	100	kg	2023-10-20	100000	Selesai
113	STOK BARANG	100	kg	2023-10-20	100000	Selesai
114	STOK BARANG	100	kg	2023-10-20	100000	Selesai
115	STOK BARANG	100	kg	2023-10-20	100000	Selesai
116	STOK BARANG	100	kg	2023-10-20	100000	Selesai
117	STOK BARANG	100	kg	2023-10-20	100000	Selesai
118	STOK BARANG	100	kg	2023-10-20	100000	Selesai
119	STOK BARANG	100	kg	2023-10-20	100000	Selesai
120	STOK BARANG	100	kg	2023-10-20	100000	Selesai

Tampilan aplikasi yang harus Anda tampilkan

► TAMPILKAN / LAKUKAN

- Dari Goods Issue Queue klik **Process Issue** pada SO Approved → klik **Issue Goods**. Stok TURUN, status jadi Fulfilled.
- Buka Stock Ledger (Kartu Stok): tunjuk baris Receipt (masuk) & Issue (keluar) — bukti jejak lengkap.

🎤 KATAKAN (baca naskah ini)

"Dan ini puncaknya. Untuk pesanan yang sudah disetujui, saya klik Issue — barang keluar, stok turun, status menjadi Selesai. Kalau saya buka Kartu Stok, terlihat seluruh jejaknya: barang masuk waktu diterima, dan keluar waktu dikeluarkan."

💡 Selip alasan — Anti-oversell (penting!) — ucapkan di momen ini

"Di balik layar, saat barang dikeluarkan, baris stok dikunci dulu (SELECT ... FOR UPDATE) supaya dua pengeluaran bersamaan tidak bisa mengambil stok yang sama. Kalau stok tidak cukup, pengeluaran ditolak dan stok tidak jadi minus. Jadi oversell dicegah secara teknis — ini salah satu syarat kritikal project ini."

Inventory & Order Management

SO-0029 APPROVED

CV Toko Elektronik Modern - Order Date: 2026-10-01 - Created by Beni

1 Draft 2 Pending Approval 3 Approved 4 Fulfilled

Sales Order & Stock Allocation Protocol: Approved Sales Orders place a hard stock reservation against the assigned Source Warehouse. Physical inventory is deducted from Product Stock only upon validated Goods Issue, recording an immutable audit entry in the Stock Ledger.

CUSTOMER: CV Toko Elektronik Modern
Send SO: 29 Approved

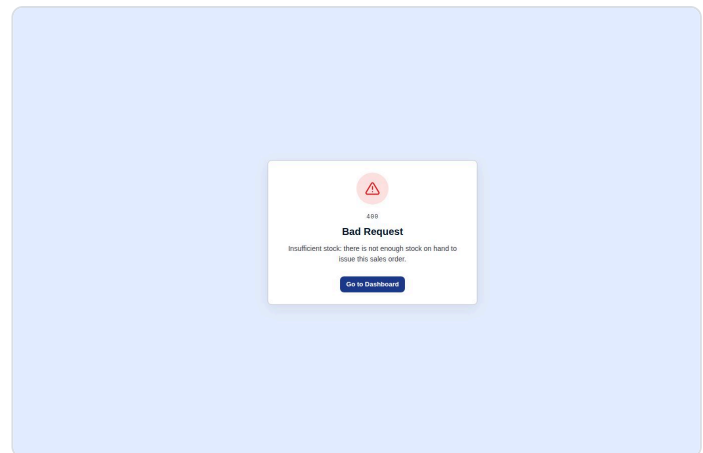
SOURCE FACILITY: Bandung Warehouse
Physical inventory deducted here upon Goods Issue.

AUDIT & CHAIN OF CUSTODY:
Order Date: 2026-10-01
Approved By: Rita (2026-10-02 00:00)
Stock Ledger Entries: 0

Line Items 2 LINES

PRODUCT	SKU	QTY	UNIT PRICE	LINE SUBTOTAL
Silicone Rubber Sheet sheet	RAM-014	8	Rp 115.000	Rp 920.000
Copper Pipe 25mm pc	RAM-015	6	Rp 115.000	Rp 690.000
Distinct SKUs: 2 - Total Quantity: 14 units				Rp 1.610.000

SO Approved dengan stok kurang



Issue ditolak: stok tidak mencukupi

Inventory & Order Management

Stock Ledger

241 movements recorded

PRODUCT SKU Please Enter Product SKU Code... **PRODUCT NAME** Please Enter Product Name...

MOVEMENT TYPE Select movement type... **WAREHOUSE** Select warehouse...

Filter **Reset**

SKU	PRODUCT NAME	WAREHOUSE	TYPE	QTY	STOCK BEFORE	AFTER
RAM-011	Carbon Steel Bar 20mm	Jakarta Warehouse	RECEIPT	+4	62	66
RAM-019	HDPE Sheet 5mm	Jakarta Warehouse	ISSUE	-10	65	55
QEC-097	Correction Tape	Bandung Warehouse	ISSUE	-6	38	24
ELEC-098	LED Monitor 24in	Jakarta Warehouse	ISSUE	-2	50	48

Stock Ledger

Inventory & Order Management

Reports

Generate operational reports, analyze historical inventory movements, and export data audits.

Report Query Parameters

REPORT TYPE Stock Valuation & Turnover Summary **DATE RANGE** 09/06/2026 to 10/05/2026 **WAREHOUSE LOCATION** All Warehouses (National)

Total Stock Valuation
Rp 1.146.877.000,00
+Rp 1.146.877.000,00 vs end of last month

Active SKU Catalog
100 Items
13.0% moved in range
1 low - 0 out of stock

Inbound / Outbound
782 In / 69 Out
+713 Net net units moved in range

Report Visual Analytics & Trends
Visual breakdown for Stock Valuation & Turnover (Sep 06, 2026 — Oct 05, 2026)
Fastest Turnover: Office Supplies • Highest Capital: Electronics (Rp 479.355.000,00)

Trend Overview (6-Month) **Category Distribution** **Warehouse Allocation**

Reports + Export Stock Ledger

► TAMPILKAN / LAKUKAN

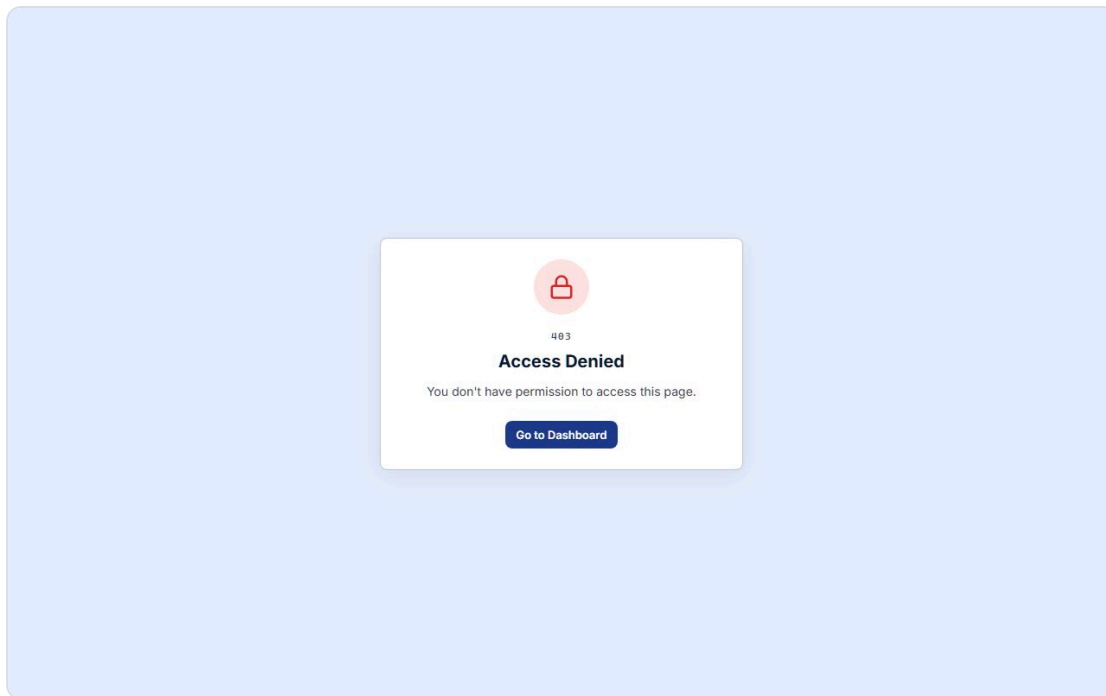
- Dari antrean Goods Issue, pilih SO yang stok gudang asalnya **tidak cukup** (bila ada di data demo) → **Issue Goods** → ditolak dengan pesan stok tidak mencukupi; status tetap Approved dan stok tidak berubah.
- Menu **Stock Ledger**: filter SKU, nama produk, jenis pergerakan (Receipt / Issue), gudang; urutkan kolom; paging 25 baris. Cocokkan dengan Receipt dan Issue yang baru dibuat.
- Menu **Reports**: pilih jenis laporan, rentang tanggal, gudang, kategori → Generate → tunjuk analitik; klik **Export Stock Ledger** → CSV.
- Tunjuk bahwa Warehouse tidak bisa mengeksport SO/PO (tidak punya izin laporan pesanan).
- Menu akun → **My Profile** → ubah nama/email → **Logout**.

🔪 KATAKAN (baca naskah ini)

"Kalau stok tidak cukup, sistem menolak pengeluaran dan tidak ada yang berubah — jadi tidak mungkin stok menjadi minus. Kartu Stok lalu memberi saya alat untuk mencocokkan barang fisik dengan catatan sistem."

💡 Selip alasan — Rekonsiliasi — ucapkan di momen ini

"Semua perubahan stok hanya lewat penerimaan dan pengeluaran yang transaksional; tidak ada fitur mengedit angka stok secara langsung. Karena itu Kartu Stok bisa dipercaya sebagai



Tampilan aplikasi yang harus Anda tampilkan

► TAMPILKAN / LAKUKAN

- Sebagai Sales atau Warehouse, ketik langsung URL menu terlarang `/users` . Muncul halaman 403 ("You don't have permission to access this page.").
- Variasi lain: Sales buka `/purchase-orders` atau `/stock-ledger` → 403.

🎤 KATAKAN (baca naskah ini)

"Terakhir, bukti keamanannya. Kalau role yang tidak berhak memaksa membuka menu terlarang dengan mengetik alamatnya langsung, dia tetap ditolak dengan halaman 403. Jadi keamanan bukan sekadar menyembunyikan menu — dicek sungguhan di server."

💡 Selip alasan — Keamanan server — ucapkan di momen ini

"Menyembunyikan tombol di tampilan bukan otorisasi; pengecekan sesungguhnya ada di server pada setiap permintaan." *Catatan jujur: tidak semua halaman daftar dijaga permission — lihat Lampiran F.*

Langkah bicara / persiapan — tidak perlu layar tertentu

► TAMPILKAN / LAKUKAN

- Rangkum perjalanan: Sales buat → Admin setuju → Warehouse keluarkan. Siap masuk tanya jawab.

KATAKAN (baca naskah ini)

"Jadi tadi satu pesanan berjalan lengkap: Sales membuatnya, Admin menyetujuinya, Warehouse mengeluarkan barangnya — dan di setiap langkah, aplikasi menjaga siapa yang berhak, mencatat jejaknya, dan memastikan stok tidak pernah minus. Itulah inti dari Inventory & Order Management System ini. Saya siap untuk pertanyaan."

Jika ditanya "kenapa Redis DAN Memcached" — ucapkan di momen ini

Jujur: Redis untuk sesi (butuh TTL) dan cache izin role, Memcached untuk cache baca produk & aturan validasi file. Keduanya sebenarnya tidak wajib — saya pakai untuk mendemonstrasikan externalized session & cache-aside, dengan peran yang dipisah jelas (ADR-005). Lihat Lampiran A dan B.

Menjabarkan kalimat di Langkah 1: *"Kalau Redis kebetulan mati, aplikasi otomatis beralih ke penyimpanan file, jadi login tetap jalan."* dan *"password dicek dengan bcrypt, tidak pernah disimpan polos."* Sumber: `app/Core/SessionManager.php`, `app/Service/AuthService.php`, `app/Service/UserService.php`, ADR-005. Fallback juga diuji langsung saat penyusunan dokumen ini (lihat A.3).

A.1 Apa itu "sesi" dan kenapa di Redis?

- Setelah login berhasil, server menyimpan `user_id`, `role`, `user_name`, dan `last_activity` di sesi PHP. Browser hanya memegang *cookie* berisi ID sesi acak (`iom_session`) — bukan data pengguna.
- Penyimpanan sesi default PHP adalah file di server. Aplikasi ini menggantinya ke **Redis** (`session.save_handler=redis`, alamat `tcp://redis:6379?database=0`) agar: (1) sesi kedaluwarsa otomatis oleh Redis (TTL), (2) cepat dibaca di tiap halaman karena di memori, (3) tidak membebani MySQL, dan (4) tidak terikat satu server/kontainer.
- Redis dipakai untuk sesi (database 0) dan cache izin role (database 1, Lampiran B). Data bisnis (termasuk stok) tidak pernah disimpan di Redis; sesi bisa hilang tanpa merusak data — dampaknya cuma pengguna harus login ulang (ADR-005).
- Masa berlaku: `SESSION_LIFETIME = 3600` detik (default di `docker-compose.yaml`). Selain cookie, server juga memeriksa **idle timeout**: bila selisih `last_activity` dengan sekarang melebihi lifetime, sesi dihancurkan dan pengguna dianggap belum login (`AuthService::currentUser()`).
- Cookie sesi diamankan: `HttpOnly` (tidak terbaca JavaScript), `SameSite=Lax`, dan flag `Secure` aktif di produksi bila lewat HTTPS (termasuk di balik reverse proxy Caddy via header `X-Forwarded-Proto`).
- Saat login ID sesi diganti (`session_regenerate_id(true)`) untuk mencegah *session fixation*, dan token CSRF dibuat baru.

A.2 Bagaimana fallback "Redis mati → file" bekerja

1. Setiap request, `SessionManager::start()` melakukan **probe TCP** singkat ke Redis (`fsockopen`, timeout 0,5 detik). Ini hanya cek "port terbuka?", bukan handshake Redis penuh.
2. Bila terjangkau → sesi disimpan di Redis. Bila tidak → aplikasi menulis peringatan ke error log (`"Redis unreachable ... falling back to file sessions"`) dan memakai `session.save_handler=files`.
3. Tujuannya (kode ditandai C-02): tanpa fallback, handler Redis yang mati membuat **setiap** request gagal atau menggantung, sehingga tak seorang pun bisa login. Dengan fallback, halaman login tetap berfungsi, pengguna bisa masuk, dan aplikasi tetap dipakai.

Batasan yang perlu diketahui (disimpulkan dari logika kode — inferensi)

Pilihan handler dibuat **per request** dan sesi tidak dipindahkan antar penyimpanan. Akibatnya: pengguna yang sudah login lewat Redis akan terlihat "logout" saat Redis mati (sesinya ada di Redis, sementara request baru membaca file), dan harus login ulang; login baru pada masa gangguan tersimpan di file lokal kontainer dan tidak terlihat lagi setelah Redis pulih. Probe hanya mengecek TCP, jadi Redis yang port-nya terbuka tetapi macet tidak terdeteksi. Jawaban jujur saat ditanya: "login tetap bisa, tetapi sesi lama tidak ikut pindah."

A.3 Bukti verifikasi

Dijalankan saat penyusunan dokumen: `SessionManager` diarahkan ke Redis palsu (`127.0.0.1:6399`). Hasil: log "falling back to file sessions", `session.save_handler=files`, sesi aktif dan data sesi dapat ditulis/dibaca. Pada mode normal (Redis hidup) handler = `redis`. Untuk demo langsung: `docker compose stop redis` lalu login → tetap berhasil; `docker compose start redis` untuk memulihkan.

A.4 Password: bcrypt, tidak pernah polos

- **Menyimpan:** saat Admin membuat user atau mengganti password, aplikasi memanggil `password_hash($password, PASSWORD_BCRYPT) ( UserService )`. Yang masuk kolom `users.password_hash` hanyalah hash berawalan `$2y$10$...` (60 karakter; `$2y$` = bcrypt PHP, `10` = cost). Password asli tidak disimpan di database, sesi, maupun log. Data seed juga berisi hash, bukan teks polos.
- **Mengecek:** saat login, `password_verify($password, $user->passwordHash)` membandingkan input dengan hash. Hash tidak dapat "dibalik" ke password; bcrypt memakai *salt* acak per password (tertanam dalam hash) sehingga dua user dengan password sama punya hash berbeda, dan sengaja lambat (cost $10 = 2^{10}$ putaran) agar sulit ditebak massal.
- Akun nonaktif, email tak dikenal, dan password salah menghasilkan pesan yang sama (anti user-enumeration); percobaan gagal dicatat sebagai `login_failed`.
- Aturan password: minimal 6 karakter saat dibuat Admin; email unik. Tidak ada ganti password mandiri — Admin menggantinya lewat Users → Edit (My Profile hanya nama & email).

Koreksi penting

Kalimat di versi sebelumnya — "sebagian data dashboard yang sering dibaca & jarang berubah dipercepat dengan cache Memcached" — **tidak sesuai kode**. `DashboardService` dan `SalesDashboardService` tidak memakai cache sama sekali; seluruh angka dashboard (nilai inventori, low stock, jumlah PO/SO per status, antrean) dibaca langsung dari MySQL tiap kali halaman dibuka. Kalimat itu sudah diganti di Langkah 2.

Hanya **dua** jenis data yang di-cache di Memcached (pola cache-aside lewat `CacheService`; sumber: `ProductService`, `FileValidationService`, ADR-005). Cache izin role ada di Redis (lihat kotak di bawah tabel):

Data (key)	Isi & dipakai di mana	TTL	Invalidasi
<code>product:<SKU></code> (SKU huruf besar)	Satu produk hasil pencarian per SKU (<code>ProductService::findBySku</code>). Dipakai mis. oleh endpoint <code>GET /api/products/{sku}/availability</code> . Hanya data produk — bukan stok .	300 detik (5 menit)	Dihapus (<code>delete</code>) saat produk diubah, dinonaktifkan atau diaktifkan (<code>invalidateSkuCache</code>); bila SKU berubah, key lama dan baru dihapus.
<code>file_validation.rules</code>	Aturan tipe gambar yang diizinkan saat upload produk (ekstensi, MIME, byte header/footer) dari tabel <code>file_validation_rules</code> .	3600 detik (1 jam)	Hanya TTL. Bila hasil kosong, upload ditolak (bukan diizinkan semua).

Cache izin role ada di Redis, bukan Memcached

`role_permissions.grouped` (seluruh tabel izin role: menu sidebar & guard `requirePermission()`) disimpan lewat `RedisCacheService` di Redis **database 1** (sesi di database 0), key berawalan `cache:`, TTL 3600 detik (1 jam). Tidak ada invalidasi manual — hanya kedaluwarsa TTL; perubahan isi `role_permissions` berlaku paling lama ± 1 jam kemudian (atau segera bila key dihapus / database 1 di-flush). Bila Redis mati, `RedisCacheService` menjadi no-op dan izin dibaca langsung dari MySQL, sehingga login dan menu tetap berfungsi.

Yang sengaja TIDAK di-cache

- **Stok** (`product_stocks`) dan ketersediaan — selalu dibaca langsung dari MySQL; penjaga konsistensinya adalah transaksi + `SELECT ... FOR UPDATE` (ADR-002), bukan cache. Alasannya agar syarat ARCH-02 (anti-oversell) tidak pernah bergantung pada data usang.
- Dashboard, daftar produk/SO/PO, laporan, Stock Ledger — semuanya langsung dari database.

Perilaku bila Memcached mati

`CacheService` mengecek koneksi saat dibuat (timeout 500 ms). Jika ekstensi/ server tidak tersedia, ia menandai diri "tidak tersedia" dan semua `get` mengembalikan kosong, `set/delete` tidak melakukan apa-apa — aplikasi tetap berfungsi penuh, hanya membaca dari MySQL. Cache bukan sumber kebenaran dan tidak pernah write-through.

Redis vs Memcached — ringkas untuk tanya-jawab

Redis = sesi login (butuh TTL, hilang tak merusak data). Memcached = cache baca produk & aturan validasi file; cache izin role juga di Redis (database terpisah dari sesi). Semuanya bisa dibangun ulang dari MySQL. Dipisah agar kelak tak ada yang "sekalian" menyimpan stok di store yang juga menjaga login.

Admin memegang semua permission key di `role_permissions`. Centang tiap butir saat dites. Format: **Lakukan → Hasil yang diharapkan**. Sumber: `config/routes.php`, `views/layouts/main.php`, controller & view terkait.

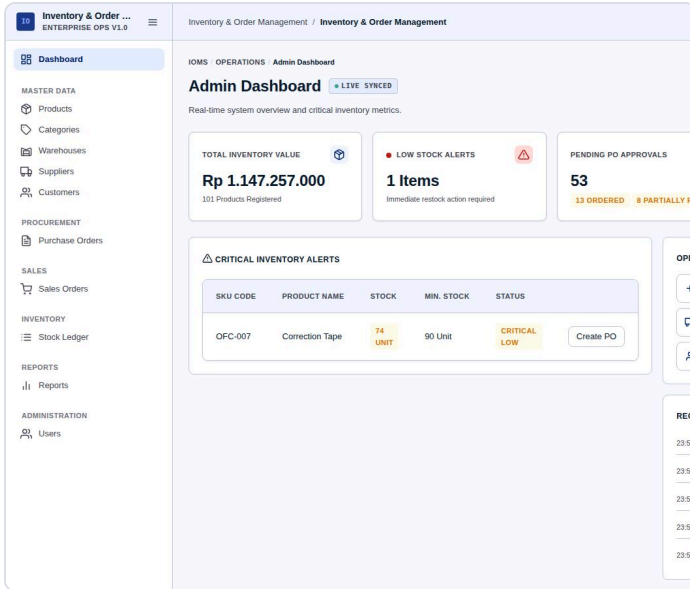
C.1 Peta menu & fungsinya

Menu (URL)	Untuk apa
Dashboard <code>/dashboard</code>	Ringkasan live: total nilai inventori & jumlah produk, produk di bawah reorder point, PO per status, SO per status (semua Sales), transaksi terbaru (5 PO + 5 SO), 5 aktivitas audit terakhir, Quick Actions.
Products <code>/products</code>	Katalog produk + stok per gudang; tambah/ubah/nonaktifkan produk.
Categories <code>/categories</code>	Pengelompokan produk; ekspor CSV daftar kategori.
Warehouses <code>/warehouses</code>	Lokasi gudang; detail menampilkan stok per produk di gudang itu.
Suppliers <code>/suppliers</code>	Pemasok untuk Purchase Order.
Customers <code>/customers</code>	Pelanggan untuk Sales Order.
Purchase Orders <code>/purchase-orders</code>	Pembelian ke supplier → menerima barang (menambah stok).
Sales Orders <code>/sales-orders</code>	Seluruh pesanan penjualan; approve/reject; kirim barang (mengurangi stok).
Stock Ledger <code>/stock-ledger</code>	Kartu stok: jejak audit semua pergerakan (Receipt/Issue).
Reports <code>/reports</code>	Analitik inventori + ekspor CSV (ledger, pesanan).
Users <code>/users</code>	Kelola akun dan role pengguna.
My Profile <code>/my-profile</code> (menu akun kanan atas)	Ubah nama & email sendiri; juga tombol Logout.
Bel notifikasi (header)	Peringatan stok rendah (dibuat job terjadwal tiap 15 menit); "Mark All as Read".

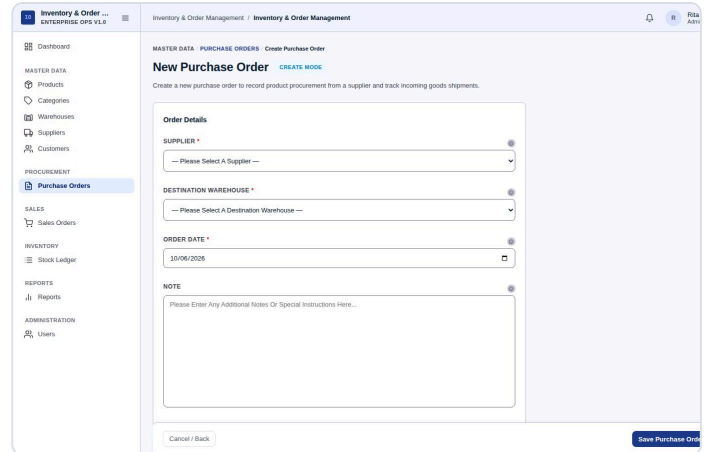
C.2 Checklist how-to

Login & Dashboard

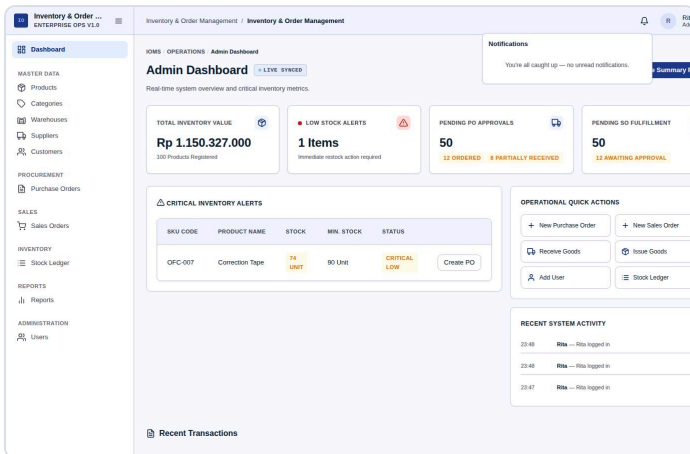
- ☐ Login `admin@example.com` → diarahkan ke `/dashboard`; sidebar menampilkan SEMUA grup (Master Data, Procurement, Sales, Inventory, Reports, Administration).
- ☐ Kartu: Total Inventory Value, Below Reorder Point, PO status, SO status ("n Awaiting Approval" bila ada) tampil; klik **Refresh** memuat ulang angka.
- ☐ Daftar Low Stock punya tombol **Create PO** → membuka form PO. Quick Actions: buat PO, buat SO, lihat PO/SO, tambah user (`/users/create`), Stock Ledger.
- ☐ Bel notifikasi: bila ada peringatan, klik **Mark All as Read** → hitungan menjadi 0.



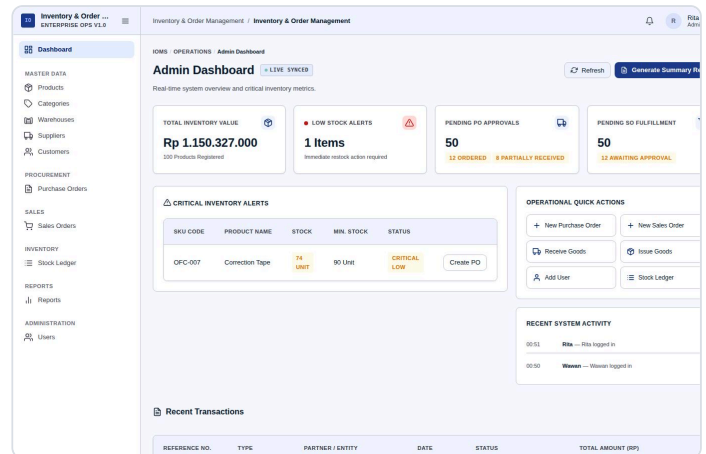
Dashboard Admin (sidebar lengkap, kartu ringkasan)



Create PO dari panel Critical Inventory Alerts



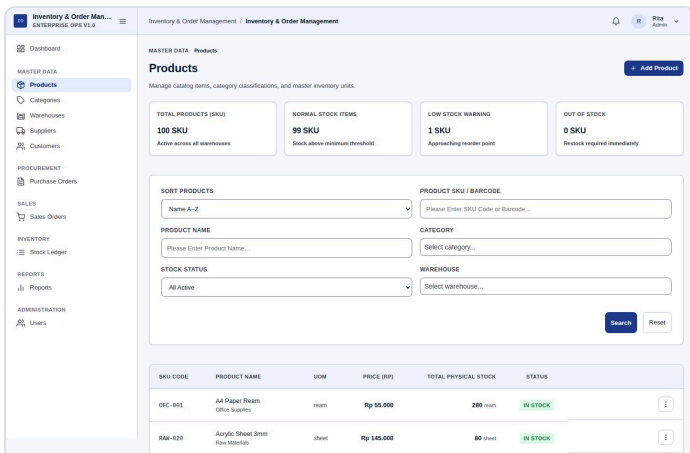
Bel notifikasi



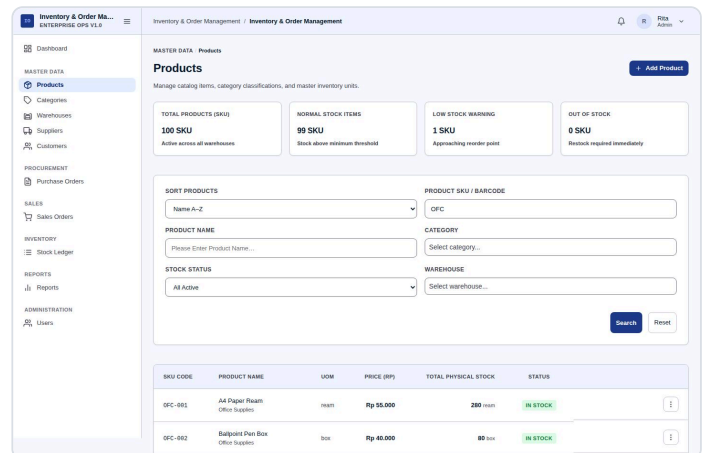
Setelah klik Refresh

Products

- Cari/filter (SKU, nama, kategori, gudang, status stok), ubah urutan & jumlah per halaman (10/25/50/100) → daftar menyesuaikan.
- Add Product: isi SKU (unik), barcode, nama (min. 3 huruf), deskripsi, kategori, unit, harga beli/jual, reorder point, gudang & qty awal (opsional), gambar JPG/PNG/WebP → tersimpan; SKU ganda ditolak dengan pesan validasi.
- Upload file bukan gambar (mis. .exe diganti nama .jpg) → ditolak (validasi ekstensi + byte signature).
- Detail produk: stok per gudang dan pergerakan terakhir. Edit → simpan. **Deactivate** → hilang dari daftar default & dropdown PO/SO baru; **Activate** mengembalikan.



Daftar Products + KPI



Filter SKU "OFC"

Form Add Product terisi

SKU ganda ditolak

File bukan gambar ditolak

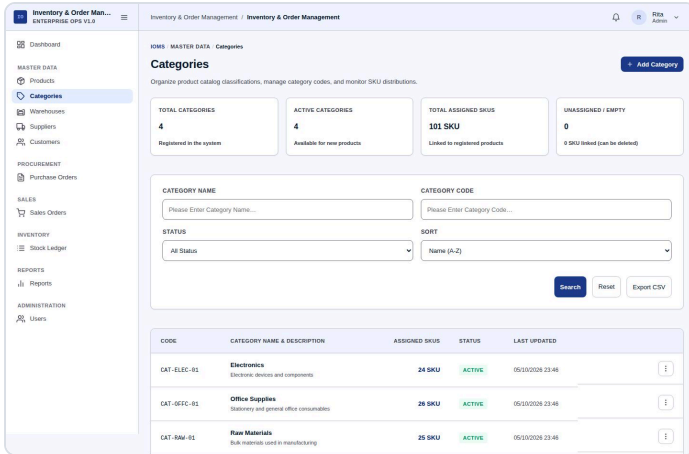
Detail produk: stok per gudang

Edit produk

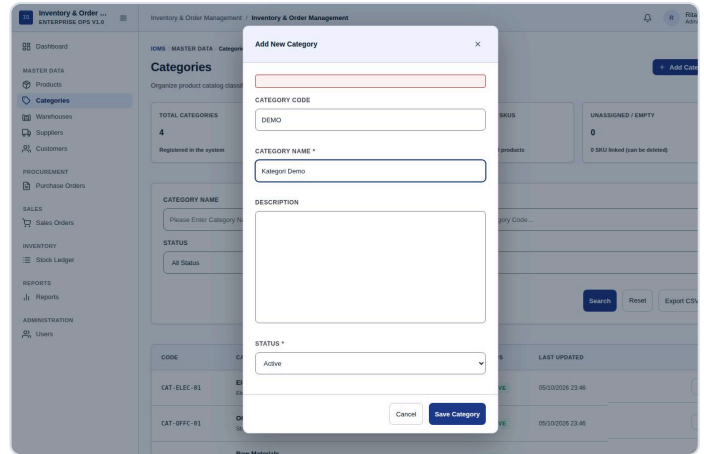
Setelah Deactivate (hilang dari daftar default)

Categories / Warehouses / Suppliers / Customers

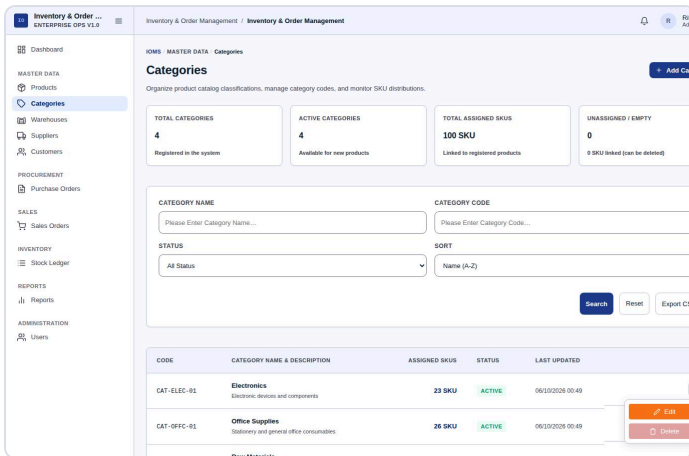
- Tiap modul: list + filter + paging; Add (Categories: kode, nama, deskripsi · Warehouses: kode, nama, lokasi · Suppliers: nama, kontak, email · Customers: nama, kontak, email, telepon) → tersimpan.
- Edit & Deactivate/Activate pada tiap modul → status berubah; data nonaktif tak muncul di dropdown transaksi baru.
- Categories saja: **Export CSV** (mengikuti filter) dan aksi **hapus sungguhan**; hapus kategori yang masih dipakai produk → ditolak.
- Warehouses → detail: jumlah produk & total qty di gudang itu, tabel stok per produk dengan paging.



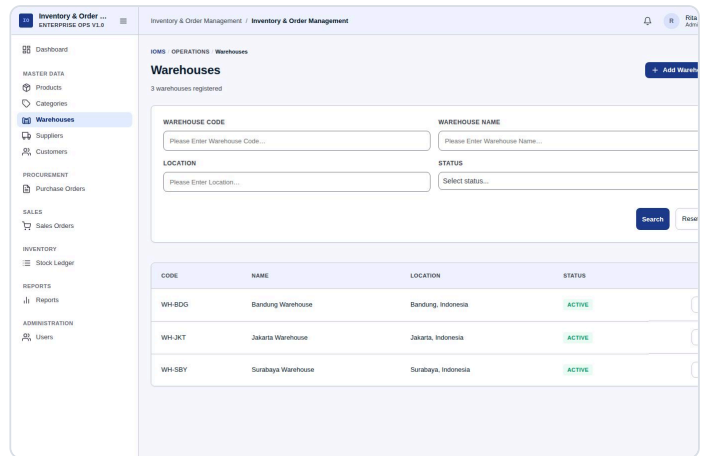
Categories



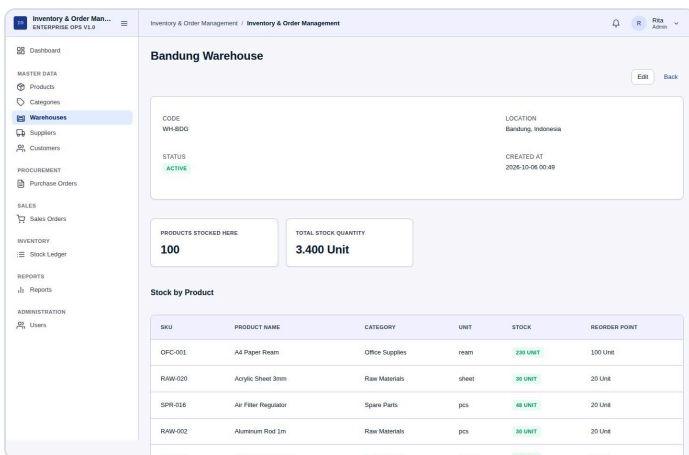
Dialog Add New Category



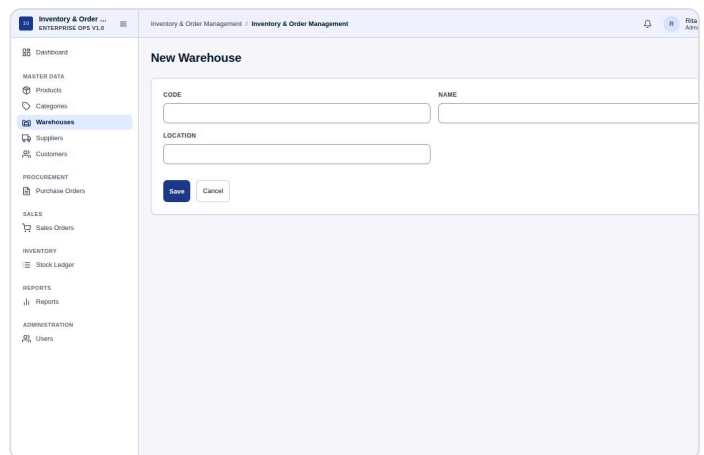
Menu baris: Delete nonaktif untuk kategori yang dipakai produk



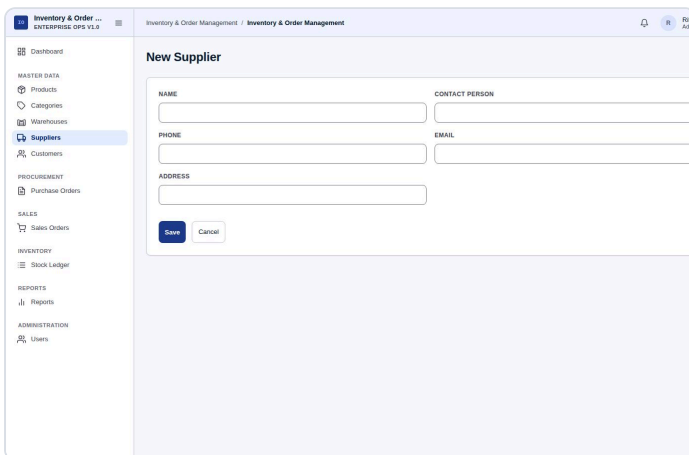
Warehouses



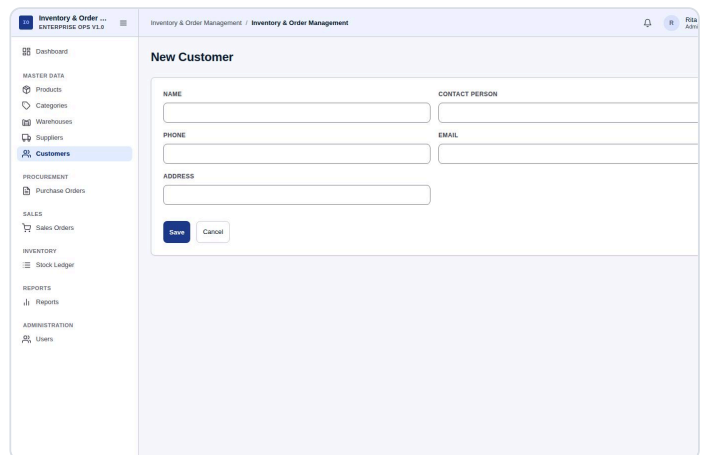
Detail gudang



Form Warehouse



Form Supplier



Form Customer

Purchase Orders

- ❑ Create PO: pilih supplier, gudang tujuan, tanggal, catatan, baris produk (qty ≥ 1 , harga ≥ 0) → status **Draft**.

- ❑ Detail Draft → **Submit Order** → **Ordered** (hanya Admin). Stepper status Draft→Ordered→PartiallyReceived→Received tampil.
- ❑ **Receive Goods** → isi qty per baris (boleh sebagian; tidak boleh melebihi sisa) → stok gudang tujuan naik, baris Receipt muncul di Stock Ledger, status PartiallyReceived/Received.
- ❑ **Cancel Order** hanya muncul/diterima sebelum ada penerimaan barang; PO yang sudah menerima barang tak bisa dibatalkan.

Daftar PO + filter

Form Create PO

PO Draft (Submit Order)

PO Ordered (Receive Goods, Cancel Order)

PO Cancelled

Sales Orders

- ❑ List menampilkan SO semua Sales; filter status, nomor SO, customer, gudang, urutan.
- ❑ Create/Edit Draft (Admin boleh mengedit Draft siapa pun) → Submit for Approval → **Approve** (tercatat approved_by & waktu) atau **Reject** dengan alasan opsional → status Cancelled.
- ❑ SO Approved → **Issue Goods** → stok gudang asal turun, baris Issue di ledger, status Fulfilled; stok tidak cukup → ditolak, tidak ada perubahan.
- ❑ **Cancel Order** pada SO apa pun yang belum Fulfilled; SO Fulfilled tidak bisa dibatalkan siapa pun.
- ❑ Admin boleh menyetujui SO buatannya sendiri (larangan self-approval hanya untuk Sales).

Inventory & Order ...
ENTERPRISE OPS V1.0

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

- Products
- Categories
- Warehouses
- Suppliers
- Customers

PROCUREMENT

- Purchase Orders

SALES

- Sales Orders

INVENTORY

- Stock Ledger

REPORTS

- Reports

ADMINISTRATION

- Users

IOMS SALES ORDERS SO-0020

Sales Orders

50 orders total

ORDER NUMBER

Please Enter Order Number...

CUSTOMER NAME

Please Enter Customer Name...

STATUS

Select status...

WAREHOUSE

Select warehouse...

ID	Date	CUSTOMER	WAREHOUSE	STATUS
#20	2026-10-06	PT Retail Maju Bersama	Bandung Warehouse	PENDING APPROVAL
#15	2026-10-06	UD Sumber Rejeki	Jakarta Warehouse	PENDING APPROVAL
#11	2026-10-06	PT Industri Manufaktur Sejaht	Jakarta Warehouse	PENDING APPROVAL
#8	2026-10-06	CV Gudang Kita	Jakarta Warehouse	DRAFT
#7	2026-10-06	PT Industri Manufaktur Sejaht	Bandung Warehouse	DRAFT

Daftar SO semua Sales

Inventory & Order Man...
ENTERPRISE OPS V1.0

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

- Products
- Categories
- Warehouses
- Suppliers
- Customers

PROCUREMENT

- Purchase Orders

SALES

- Sales Orders

INVENTORY

- Stock Ledger

REPORTS

- Reports

ADMINISTRATION

- Users

IOMS SALES ORDERS SO-0020

PENDING APPROVAL

PT Retail Maju Bersama Order Date: 2026-10-06 Created by Grace

1 Draft

2 Pending Approval

3 Approved

4 Fulfilled

Sales Order & Stock Allocation Protocol. Approved Sales Orders place a hard stock reservation against the assigned Source Warehouse. Physical inventory is deducted from ProductStock only upon validated Goods Issue, recording an immutabile audit entry in the Stock Ledger.

CUSTOMER

PT Retail Maju Bersama

Seed SO 20 Pind

SOURCE FACILITY

Bandung Warehouse

Physical inventory deducted here upon Goods Issue.

AUDIT & CHAIN OF CUSTODY

Order Date: 2026-10-06

Stock Ledger Entries: 0

Line Items 1 LINE

PRODUCT	SKU	QTY	UNIT PRICE	LINE SUBTOTAL
Paper Cutler A4 jps	0FC-817	25	Rp 42.000	Rp 1.050.000
Distinct SKUs: 1 - Total Quantity: 25 units				Rp 1.050.000

Approve

Reason (optional)

Reject

Cancel Order

Back

SO Pending: Approve / Reject

Inventory & Order Man...
ENTERPRISE OPS V1.0

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

- Products
- Categories
- Warehouses
- Suppliers
- Customers

PROCUREMENT

- Purchase Orders

SALES

- Sales Orders

INVENTORY

- Stock Ledger

REPORTS

- Reports

ADMINISTRATION

- Users

IOMS SALES ORDERS SO-0020

APPROVED

PT Retail Maju Bersama Order Date: 2026-10-06 Created by Grace

1 Draft

2 Pending Approval

3 Approved

4 Fulfilled

Sales Order & Stock Allocation Protocol. Approved Sales Orders place a hard stock reservation against the assigned Source Warehouse. Physical inventory is deducted from ProductStock only upon validated Goods Issue, recording an immutabile audit entry in the Stock Ledger.

CUSTOMER

PT Retail Maju Bersama

Seed SO 20 Pind

SOURCE FACILITY

Bandung Warehouse

Physical inventory deducted here upon Goods Issue.

AUDIT & CHAIN OF CUSTODY

Order Date: 2026-10-06

Approved By: Risa (2026-10-06 00:53)

Stock Ledger Entries: 0

Line Items 1 LINE

PRODUCT	SKU	QTY	UNIT PRICE	LINE SUBTOTAL
Paper Cutler A4 jps	0FC-817	25	Rp 42.000	Rp 1.050.000
Distinct SKUs: 1 - Total Quantity: 25 units				Rp 1.050.000

Issue Goods

Cancel Order

Back

Setelah Approve (Issue Goods, Cancel Order)

Inventory & Order Man...
ENTERPRISE OPS V1.0

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

- Products
- Categories
- Warehouses
- Suppliers
- Customers

PROCUREMENT

- Purchase Orders

SALES

- Sales Orders

INVENTORY

- Stock Ledger

REPORTS

- Reports

ADMINISTRATION

- Users

IOMS SALES ORDERS SO-0020

FULFILLED

PT Retail Maju Bersama Order Date: 2026-10-06 Created by Grace

1 Draft

2 Pending Approval

3 Approved

4 Fulfilled

Sales Order & Stock Allocation Protocol. Approved Sales Orders place a hard stock reservation against the assigned Source Warehouse. Physical inventory is deducted from ProductStock only upon validated Goods Issue, recording an immutabile audit entry in the Stock Ledger.

CUSTOMER

PT Retail Maju Bersama

Seed SO 20 Pind

SOURCE FACILITY

Bandung Warehouse

Physical inventory deducted here upon Goods Issue.

AUDIT & CHAIN OF CUSTODY

Order Date: 2026-10-06

Approved By: Risa (2026-10-06 00:53)

Issued By: Risa (2026-10-06 00:53)

Stock Ledger Entries: 1

Line Items 1 LINE

PRODUCT	SKU	QTY	UNIT PRICE	LINE SUBTOTAL
Paper Cutler A4 jps	0FC-817	25	Rp 42.000	Rp 1.050.000
Distinct SKUs: 1 - Total Quantity: 25 units				Rp 1.050.000

Audit Log

Setelah Issue Goods: Fulfilled

Inventory & Order Man...
ENTERPRISE OPS V1.0

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

- Products
- Categories
- Warehouses
- Suppliers
- Customers

PROCUREMENT

- Purchase Orders

SALES

- Sales Orders

INVENTORY

- Stock Ledger

REPORTS

- Reports

ADMINISTRATION

- Users

IOMS SALES ORDERS SO-0020

CANCELLED

PT Retail Maju Bersama Order Date: 2026-10-06 Created by Grace

1 Draft

2 Pending Approval

3 Approved

4 Fulfilled

5 Cancelled

Sales Order & Stock Allocation Protocol. Approved Sales Orders place a hard stock reservation against the assigned Source Warehouse. Physical inventory is deducted from ProductStock only upon validated Goods Issue, recording an immutabile audit entry in the Stock Ledger.

CUSTOMER

PT Retail Maju Bersama

Seed SO 20 Pind

SOURCE FACILITY

Bandung Warehouse

Physical inventory deducted here upon Goods Issue.

AUDIT & CHAIN OF CUSTODY

Order Date: 2026-10-06

Stock Ledger Entries: 0

Line Items 1 LINE

PRODUCT	SKU	QTY	UNIT PRICE	LINE SUBTOTAL
Paper Cutler A4 jps	0FC-817	25	Rp 42.000	Rp 1.050.000
Distinct SKUs: 1 - Total Quantity: 25 units				Rp 1.050.000

Back

Setelah Reject: Cancelled + alasan

Inventory & Order Man...
ENTERPRISE OPS V1.0

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

- Products
- Categories
- Warehouses
- Suppliers
- Customers

PROCUREMENT

- Purchase Orders

SALES

- Sales Orders

INVENTORY

- Stock Ledger

REPORTS

- Reports

ADMINISTRATION

- Users

IOMS SALES ORDERS SO-0020

PENDING APPROVAL

UD Sumber Rejeki Order Date: 2026-10-06 Created by Grace

1 Draft

2 Pending Approval

3 Approved

4 Fulfilled

Sales Order & Stock Allocation Protocol. Approved Sales Orders place a hard stock reservation against the assigned Source Warehouse. Physical inventory is deducted from ProductStock only upon validated Goods Issue, recording an immutabile audit entry in the Stock Ledger.

CUSTOMER

UD Sumber Rejeki

Seed SO 15 Pind

SOURCE FACILITY

Jakarta Warehouse

Physical inventory deducted here upon Goods Issue.

AUDIT & CHAIN OF CUSTODY

Order Date: 2026-10-06

Stock Ledger Entries: 0

Line Items 2 LINES

PRODUCT	SKU	QTY	UNIT PRICE	LINE SUBTOTAL
Ball Bearing 6005 jps	SPR-812	15	Rp 48.000	Rp 720.000
Oil Seal 25x45x7 jps	SPR-813	20	Rp 25.000	Rp 500.000
Distinct SKUs: 2 - Total Quantity: 35 units				Rp 1.220.000

Approve

Reason (optional)

Reject

Cancel Order

Back

Konfirmasi Cancel Order

Inventory & Order Man...
ENTERPRISE OPS V1.0

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

- Products
- Categories
- Warehouses
- Suppliers
- Customers

PROCUREMENT

- Purchase Orders

SALES

- Sales Orders

INVENTORY

- Stock Ledger

REPORTS

- Reports

ADMINISTRATION

- Users

IOMS SALES ORDERS SO-0015

CANCELLED

UD Sumber Rejeki Order Date: 2026-10-06 Created by Beni

1 Draft

2 Pending Approval

3 Approved

4 Fulfilled

5 Cancelled

Sales Order & Stock Allocation Protocol. Approved Sales Orders place a hard stock reservation against the assigned Source Warehouse. Physical inventory is deducted from ProductStock only upon validated Goods Issue, recording an immutabile audit entry in the Stock Ledger.

CUSTOMER

UD Sumber Rejeki

Seed SO 15 Pind

SOURCE FACILITY

Jakarta Warehouse

Physical inventory deducted here upon Goods Issue.

AUDIT & CHAIN OF CUSTODY

Order Date: 2026-10-06

Stock Ledger Entries: 0

Line Items 2 LINES

PRODUCT	SKU	QTY	UNIT PRICE	LINE SUBTOTAL
Ball Bearing 6005 jps	SPR-812	15	Rp 48.000	Rp 720.000
Oil Seal 25x45x7 jps	SPR-813	20	Rp 25.000	Rp 500.000
Distinct SKUs: 2 - Total Quantity: 35 units				Rp 1.220.000

SO dibatalkan

Inventory & Order Man...
ENTERPRISE OPS V1.0

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

- Products
- Categories
- Warehouses
- Suppliers
- Customers

PROCUREMENT

- Purchase Orders

SALES

- Sales Orders

INVENTORY

- Stock Ledger

REPORTS

- Reports

ADMINISTRATION

- Users

IOMS SALES ORDERS SO-0008

DRAFT

CV Gudang Kita Order Date: 2026-10-05 Created by Grace

1 Draft

2 Pending Approval

3 Approved

4 Fulfilled

Sales Order & Stock Allocation Protocol. Approved Sales Orders place a hard stock reservation against the assigned Source Warehouse. Physical inventory is deducted from ProductStock only upon validated Goods Issue, recording an immutabile audit entry in the Stock Ledger.

CUSTOMER

CV Gudang Kita

Seed SO 8 Draft

SOURCE FACILITY

Jakarta Warehouse

Physical inventory deducted here upon Goods Issue.

AUDIT & CHAIN OF CUSTODY

Order Date: 2026-10-05

Stock Ledger Entries: 0

Line Items 1 LINE

PRODUCT	SKU	QTY	UNIT PRICE	LINE SUBTOTAL
Gaming Mousepad XL jps	ELEC-819	5	Rp 105.000	Rp 525.000
Distinct SKUs: 1 - Total Quantity: 5 units				Rp 525.000

Edit Draft

Cancel Order

Back

Admin melihat SO Draft milik Sales

Stock Ledger & Reports

51 / 65

- ❑ Stock Ledger: filter SKU, nama produk, jenis (Receipt/Issue/Adjustment), gudang; urut kolom; 25 baris/halaman → hasil sesuai. Tidak ada tombol edit/hapus baris.
- ❑ Reports: pilih jenis laporan (Stock Valuation & Turnover, Inventory Aging, Slow-Moving & Dead Stock, Movement Ledger Audit), rentang tanggal, gudang, kategori, pencarian, urutan → **Generate**; grafik & tabel (5 baris/halaman) berubah.
- ❑ **Export Stock Ledger** (CSV, mengikuti filter) dan **Export Orders** (CSV gabungan SO semua Sales + semua PO; wajib isi tanggal mulai & akhir, mulai ≤ akhir); tombol preview menampilkan tabel contoh isi CSV.

Inventory & Order ...
ENTERPRISE OPS V1.0

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

Products

Categories

Warehouses

Suppliers

Customers

PROCUREMENT

Purchase Orders

SALES

Sales Orders

INVENTORY

Stock Ledger

REPORTS

Reports

ADMINISTRATION

Users

IOMS INVENTORY Stock Ledger

Stock Ledger

241 movements recorded

PRODUCT SKU

Please Enter Product SKU Code...

MOVEMENT TYPE

Select movement type...

Filter

Reset

PRODUCT NAME

Please Enter Product Name...

WAREHOUSE

Select warehouse...

SKU	PRODUCT NAME	WAREHOUSE	TYPE	QTY	STOCK BEFORE	AFTER
RAM-011	Carbon Steel Bar 20mm	Jakarta Warehouse	RECEIPT	+4	62	66
RAM-018	HDPE Sheet 5mm	Jakarta Warehouse	ISSUE	-10	65	55
OFC-087	Correction Tape	Bandung Warehouse	ISSUE	-6	38	24
ELEC-088	LED Monitor 24in	Jakarta Warehouse	ISSUE	-2	58	48

Stock Ledger

Inventory & Order ...
ENTERPRISE OPS V1.0

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

Products

Categories

Warehouses

Suppliers

Customers

PROCUREMENT

Purchase Orders

SALES

Sales Orders

INVENTORY

Stock Ledger

REPORTS

Reports

ADMINISTRATION

Users

IOMS INVENTORY Stock Ledger

Stock Ledger

62 movements recorded

PRODUCT SKU

OFC

MOVEMENT TYPE

Select movement type...

Filter

Reset

PRODUCT NAME

Please Enter Product Name...

WAREHOUSE

Select warehouse...

SKU	PRODUCT NAME	WAREHOUSE	TYPE	QTY	STOCK BEFORE	AFTER	REFERENCE / NOTE	DONE BY	DATE
OFC-017	Paper Cutter A4	Bandung Warehouse	ISSUE	-25	38	5	SO #20	Risa	06 Oct 2026 00:51
OFC-087	Correction Tape	Bandung Warehouse	ISSUE	-6	38	24	SO #42	Wawan	06 Oct 2026 00:49
OFC-088	File Folder Box	Bandung Warehouse	ISSUE	-10	38	28	SO #38	Wawan	06 Oct 2026 00:49
OFC-085	Whiteboard Marker Set	Bandung Warehouse	ISSUE	-6	38	24	SO #34	Wawan	06 Oct 2026 00:49

Filter SKU "OFC"

Inventory & Order Man...
ENTERPRISE OPS V1.0

Inventory & Order Management / Reports

Dashboard

MASTER DATA

Products

Categories

Warehouses

Suppliers

Customers

PROCUREMENT

Purchase Orders

SALES

Sales Orders

INVENTORY

Stock Ledger

REPORTS

Reports

ADMINISTRATION

Users

IOMS OPERATIONS Reports

Reports

Generate operational reports, analyze historical inventory movements, and export data audits.

Report Query Parameters

REPORT TYPE *
Stock Valuation & Turnover Summary

DATE RANGE *
09/06/2026 to 10/05/2026

WAREHOUSE LOCATION
All Warehouses (National)

PRODUCT CATEGORY
All Categories

Reset Filters

Apply & Generate

Total Stock Valuation

Rp 1.150.327.000,00

+Rp 3.558.327.000,00 vs end of last month

Active SKU Catalog

100 Items

12.0% moved in range 1 low - 0 out of stock

Inbound / Outbound

782 In / 59 Out

+723 Net net units moved in range

Turnover Velocity

0.1x / yr

Below Target Benchmark Target: +4.5x / yr

Report Visual Analytics & Trends

Visual breakdown for Stock Valuation & Turnover (Sep 06, 2026 — Oct 05, 2026)

Fastest Turnover: Office Supplies • Highest Capital: Electronics (Rp 479.355.000,00)

Trend Overview (6-Month)

Category Distribution

Warehouse Allocation

Valuation (Rp M)

Inbound Volume (Units)

Outbound Volume (Units)

Target Turnover SLA

1K

773

258

0

SLA

Reports: Stock Valuation

Inventory & Order ...
ENTERPRISE OPS V1.0

Inventory & Order Management / Reports

Dashboard

MASTER DATA

Products

Categories

Warehouses

Suppliers

Customers

PROCUREMENT

Purchase Orders

SALES

Sales Orders

INVENTORY

Stock Ledger

REPORTS

Reports

ADMINISTRATION

Users

IOMS OPERATIONS Reports

Reports

Generate operational reports, analyze historical inventory movements, and export data audits.

Report Query Parameters

REPORT TYPE *
Inventory Aging Analysis

DATE RANGE *
09/07/2026 to 10/06/2026

WAREHOUSE LOCATION
All Warehouses (National)

Total Stock Valuation

Rp 1.149.702.000,00

+Rp 1.149.702.000,00 vs end of last month

Active SKU Catalog

100 Items

13.0% moved in range 1 low - 0 out of stock

Inbound / Outbound

782 In / 84 Out

+698 Net net units moved in range

Report Visual Analytics & Trends

Visual breakdown for Stock Valuation & Turnover (Sep 07, 2026 — Oct 06, 2026)

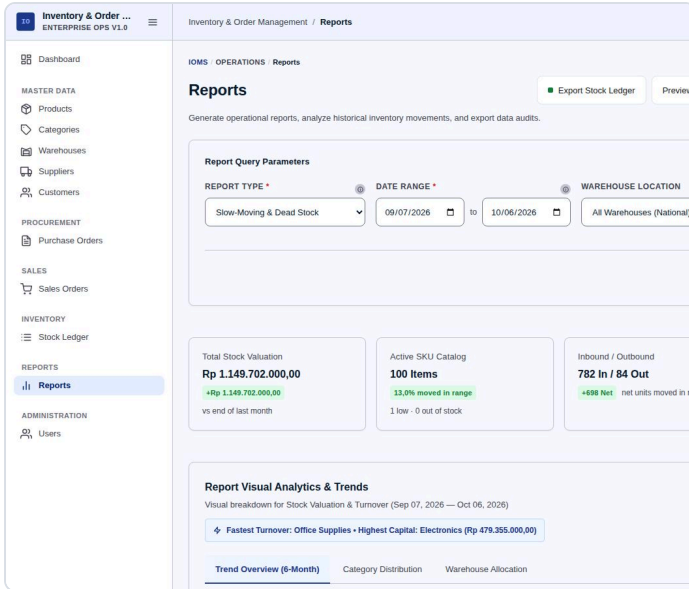
Fastest Turnover: Office Supplies • Highest Capital: Electronics (Rp 479.355.000,00)

Trend Overview (6-Month)

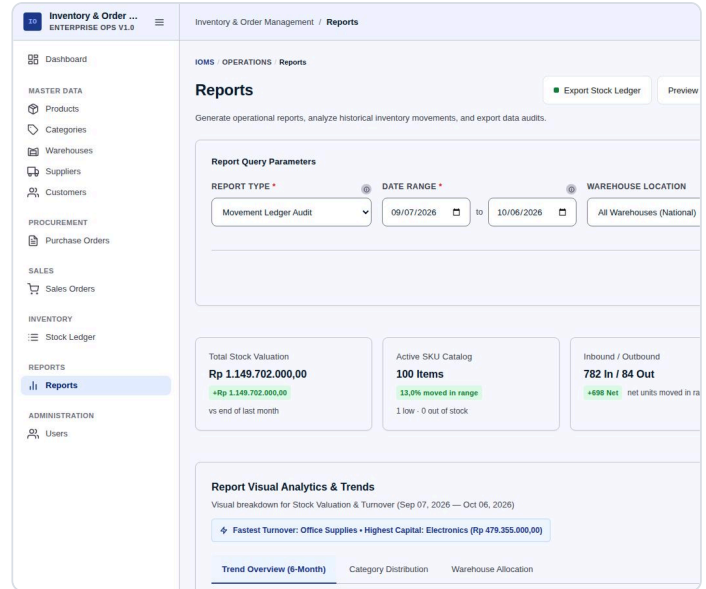
Category Distribution

Warehouse Allocation

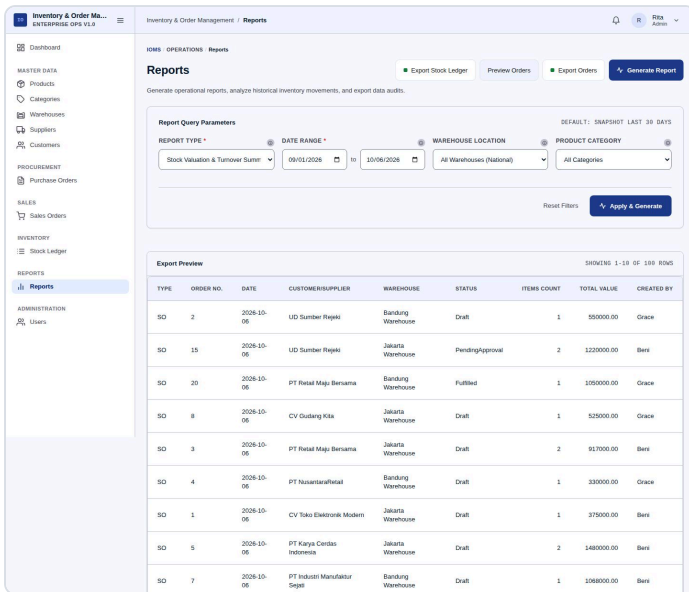
Inventory Aging Analysis



Slow-Moving & Dead Stock



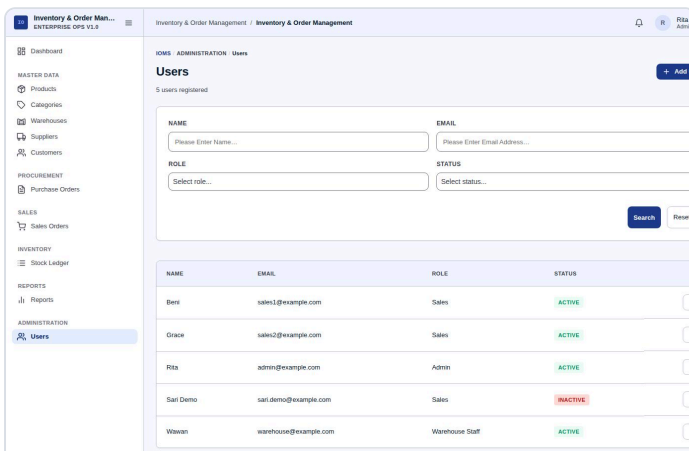
Movement Ledger Audit



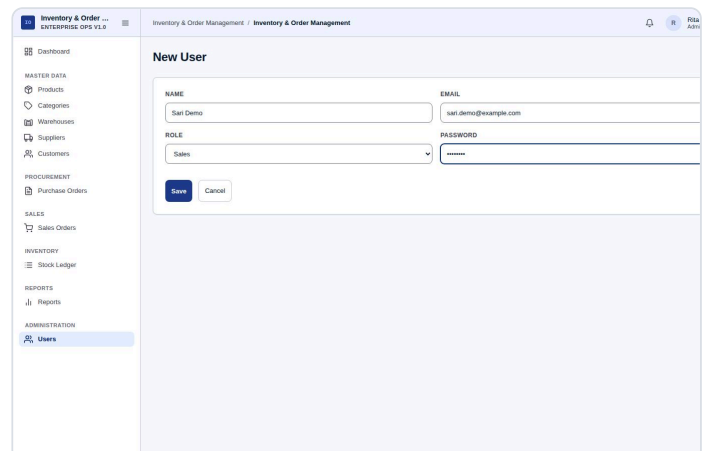
Preview Export Orders (SO semua Sales + PO)

Users & Profile

- ❑ Add User: nama, email (unik, tak peka huruf besar/kecil), role (Admin/Sales/WarehouseStaff), password (min. 6) → bisa langsung login.
- ❑ Edit user: password dikosongkan = tidak berubah. Deactivate → user itu tidak bisa login & sesi aktifnya ditolak pada request berikutnya; Activate memulihkan.
- ❑ Admin tidak bisa menonaktifkan akunnnya sendiri atau mengubah role-nya sendiri → ditolak.
- ❑ My Profile: ubah nama/email → muncul pesan sukses. Logout (menu akun) → kembali ke login.



Daftar Users



Form New User

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

- Products
- Categories
- Warehouses
- Suppliers
- Customers

PROCUREMENT

- Purchase Orders

SALES

- Sales Orders

INVENTORY

- Stock Ledger

REPORTS

- Reports

ADMINISTRATION

- Users**

New User

This email is already in use.

NAME: Sati Dika

EMAIL: sati.demo@example.com

ROLE: Sales

PASSWORD: [input]

Save Cancel

Email ganda ditolak

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

- Products
- Categories
- Warehouses
- Suppliers
- Customers

PROCUREMENT

- Purchase Orders

SALES

- Sales Orders

INVENTORY

- Stock Ledger

REPORTS

- Reports

ADMINISTRATION

- Users**

Users

4 users registered

NAME: [input] EMAIL: [input]

ROLE: [input] STATUS: [input]

Search Reset

NAME	EMAIL	ROLE	STATUS
Beni	sales1@example.com	Sales	ACTIVE
Grace	sales2@example.com	Sales	ACTIVE
Rita	admin@example.com	Admin	ACTIVE
Wawan	warehouse@example.com	Warehouse Staff	ACTIVE

View Edit

Akun Admin sendiri: tidak ada Deactivate

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

- Products
- Categories
- Warehouses
- Suppliers
- Customers

PROCUREMENT

- Purchase Orders

SALES

- Sales Orders

INVENTORY

- Stock Ledger

REPORTS

- Reports

ADMINISTRATION

- Users**

Users

5 users registered

NAME: [input] EMAIL: [input]

ROLE: [input] STATUS: [input]

Search Reset

NAME	EMAIL	ROLE	STATUS
Beni	sales1@example.com	Sales	ACTIVE
Grace	sales2@example.com	Sales	ACTIVE
Rita	admin@example.com	Admin	ACTIVE
Sati Demo	sati.demo@example.com	Sales	INACTIVE
Wawan	warehouse@example.com	Warehouse Staff	ACTIVE

User dinonaktifkan

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

- Products
- Categories
- Warehouses
- Suppliers
- Customers

PROCUREMENT

- Purchase Orders

SALES

- Sales Orders

INVENTORY

- Stock Ledger

REPORTS

- Reports

ADMINISTRATION

- Users**

My Profile

View and update your account information.

Rita Admin • Joined Oct 2025

Account Information

FULL NAME *: Rita

EMAIL ADDRESS *: admin@example.com

ROLE: Admin

Cancel / Back Save Changes

My Profile

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

- Products
- Categories
- Warehouses
- Suppliers
- Customers

PROCUREMENT

- Purchase Orders

SALES

- Sales Orders

INVENTORY

- Stock Ledger

REPORTS

- Reports

ADMINISTRATION

- Users**

My Profile

View and update your account information.

Profile updated successfully.

Rita Admin • Joined Oct 2025

Account Information

FULL NAME *: Rita

EMAIL ADDRESS *: admin@example.com

ROLE: Admin

Cancel / Back Save Changes

Profil tersimpan

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

- Products
- Categories
- Warehouses
- Suppliers
- Customers

PROCUREMENT

- Purchase Orders

SALES

- Sales Orders

INVENTORY

- Stock Ledger

REPORTS

- Reports

ADMINISTRATION

- Users**

Admin Dashboard

Real-time system overview and critical inventory metrics.

TOTAL INVENTORY VALUE: Rp 1.150.327.000

LOW STOCK ALERTS: 1 Items

PENDING PO APPROVALS: 52

PENDING SO FULFILLMENT: 50

CRITICAL INVENTORY ALERTS

SKU CODE	PRODUCT NAME	STOCK	MIN. STOCK	STATUS
OFC-007	Correction Tape	74 UNIT	90 UNIT	CRITICAL LOW

Operational Quick Actions

- New Purchase Order
- New Sales Order
- Receive Goods
- Issue Goods
- Add User
- Stock Ledger

Recent System Activity

- Rita - Requested SO #22 (Data pending and data mismatch)
- Rita - Cancelled PO #52
- Rita - Created PO #52 (Draft)
- Rita - Submitted PO #52 to supplier
- Rita - Created PO #52 (Draft)

Menu akun: My Profile & Logout

Permission Sales: `customers.view`, `sales_orders.menu`, `reports.sales_orders.view`. Tidak punya semua permission `*.manage`, `approve`, `issue`, `stock ledger`.

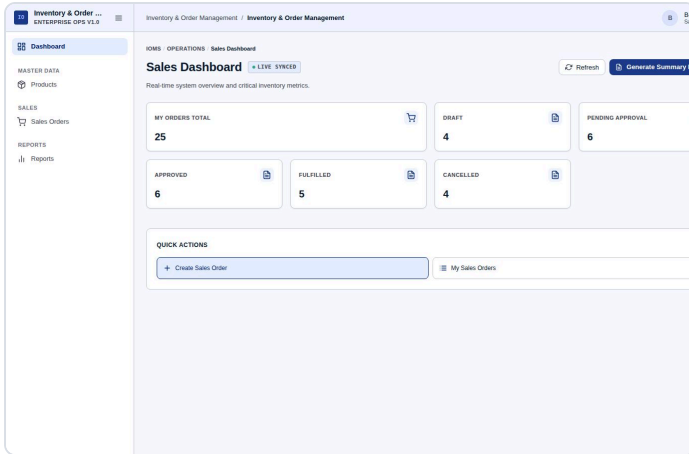
D.1 Peta menu & fungsinya

Menu (URL)	Untuk apa
Dashboard <code>/dashboard</code>	Jumlah pesanan milik sendiri : total + per status (Draft, Pending Approval, Approved, Fulfilled, Cancelled); Quick Actions "Create Sales Order" & "My Sales Orders". Tanpa bel notifikasi.
Products <code>/products</code> (read-only)	Melihat katalog, harga jual, dan stok untuk menyusun pesanan. Tanpa tombol tambah/ubah.
Sales Orders <code>/sales-orders</code>	Fitur utama: membuat, mengedit Draft, mengajukan, membatalkan, dan memantau pesanan miliknya.
Reports <code>/reports</code>	Hanya ekspor CSV pesanan miliknya (halaman versi ringkas: rentang tanggal + Export Orders).
My Profile	Ubah nama & email sendiri.

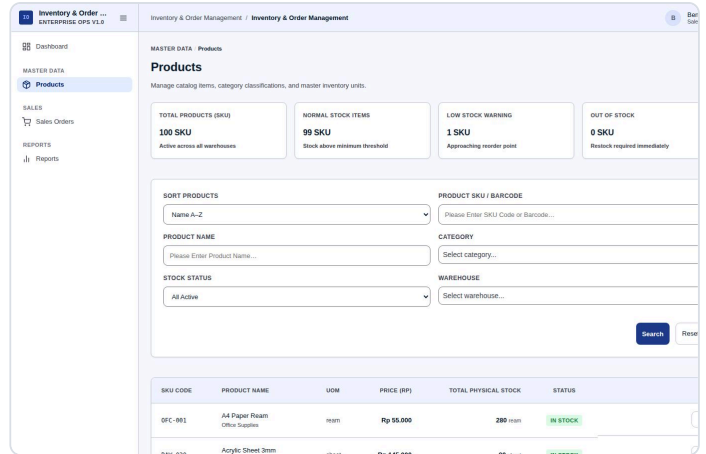
Catatan: menu **Customers tidak ada di sidebar Sales** (link hanya muncul bila punya `master_data.menu`, milik Admin). Namun Sales punya `customers.view`, jadi `/customers` bisa dibuka lewat URL (read-only). Customer dipilih lewat dropdown di form SO. Halaman `/sales-dashboard` (ringkasan dengan pemilih periode) juga ada dan dijaga `sales_orders.menu`, tetapi tidak ditemukan tautan menuju halaman itu di tampilan — **inferensi** dari pencarian view.

D.2 Checklist how-to

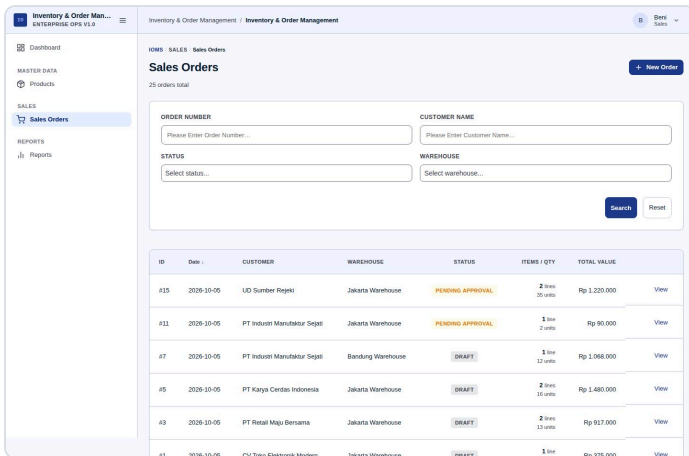
- ☐ Login `sales1@example.com` → dashboard Sales; sidebar hanya Dashboard, Products, Sales Orders, Reports.
- ☐ Products: cari/filter; tidak ada tombol Add/Edit/Deactivate. Buka `/products/create` langsung → 403.
- ☐ Sales Orders → list hanya berisi SO buatan sendiri; filter status/nomor/customer/gudang bekerja.
- ☐ Create Sales Order: pilih customer aktif, gudang asal aktif, tanggal, baris produk ($qty \geq 1$, $harga \geq 0$) → status **Draft**; stok tidak berubah.
- ☐ Simpan tanpa baris item, atau dengan $qty = 0$ → ditolak dengan pesan validasi.
- ☐ Detail Draft: tombol **Edit Draft** (ubah & simpan), **Submit for Approval**, **Cancel Order**.
- ☐ Submit → **Pending Approval**; sekarang tombol Edit/Submit hilang, tombol Approve/Reject tidak ada, Cancel Order masih ada.
- ☐ Cancel pada SO Draft/Pending milik sendiri → konfirmasi "Cancel this sales order?" → status Cancelled. Pada SO Approved/Fulfilled → tombol Cancel tidak tersedia (minta Admin).
- ☐ Setelah Admin approve: status Approved; tidak ada tombol Issue Goods untuk Sales. Setelah Warehouse issue: Fulfilled.
- ☐ Reports: pilih rentang tanggal → **Export Orders** → CSV hanya berisi pesanan Sales ini; tanggal kosong atau mulai > akhir → ditolak.
- ☐ My Profile: ubah nama/email → sukses. Logout.



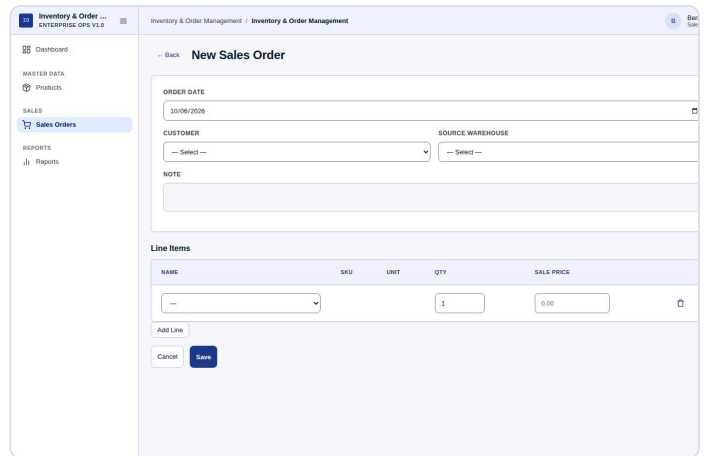
Dashboard Sales (sidebar 4 menu)



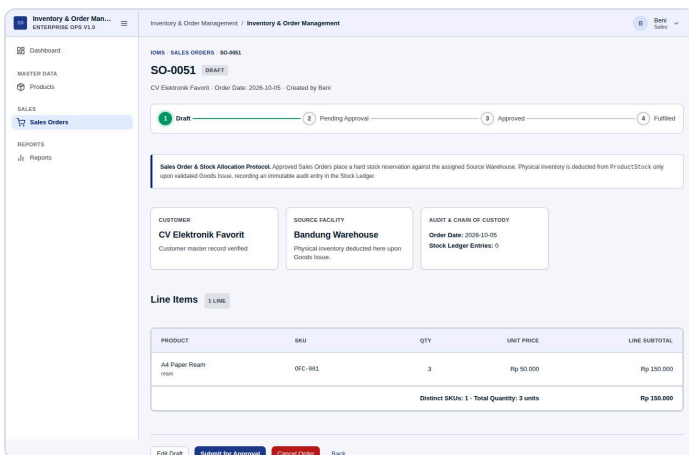
Products read-only



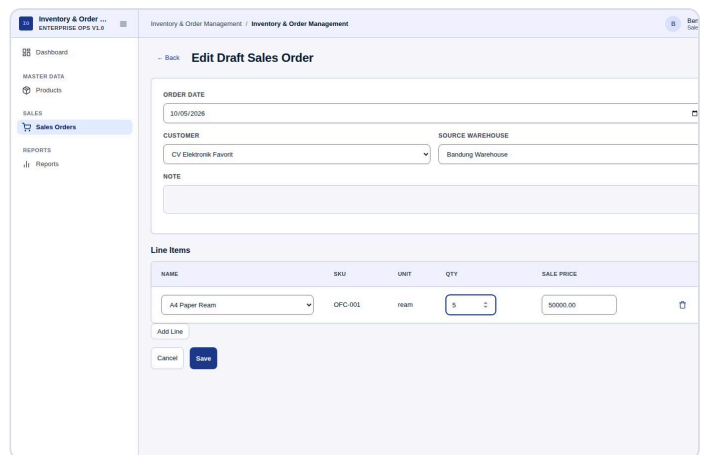
Daftar SO milik sendiri



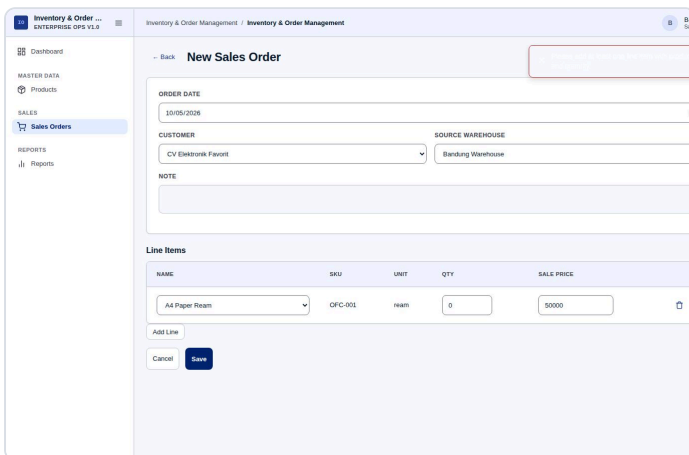
Form Create SO



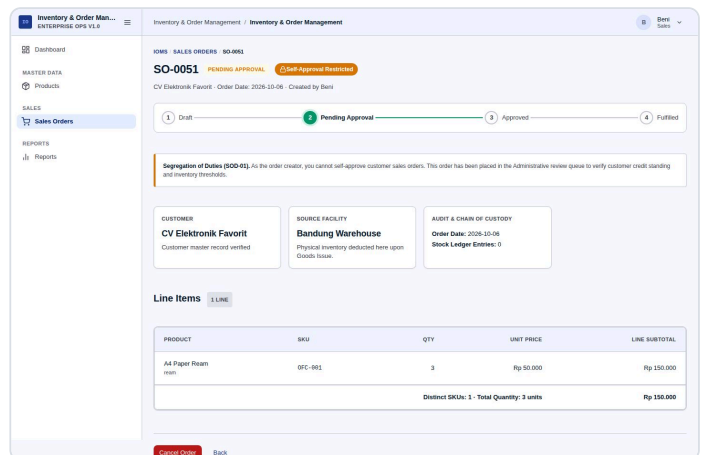
Detail Draft: Edit / Submit / Cancel



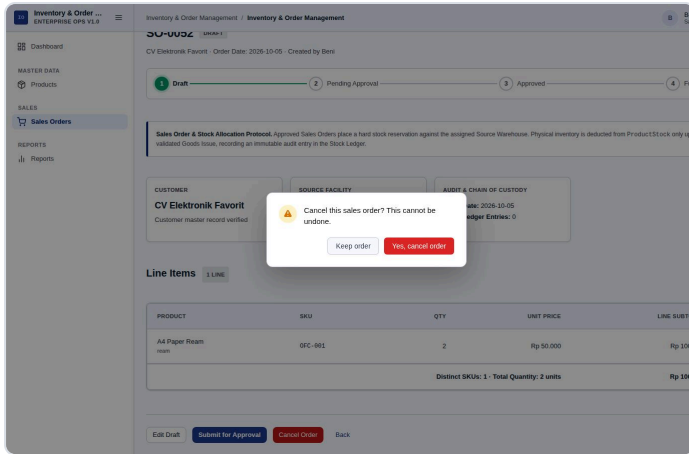
Edit Draft



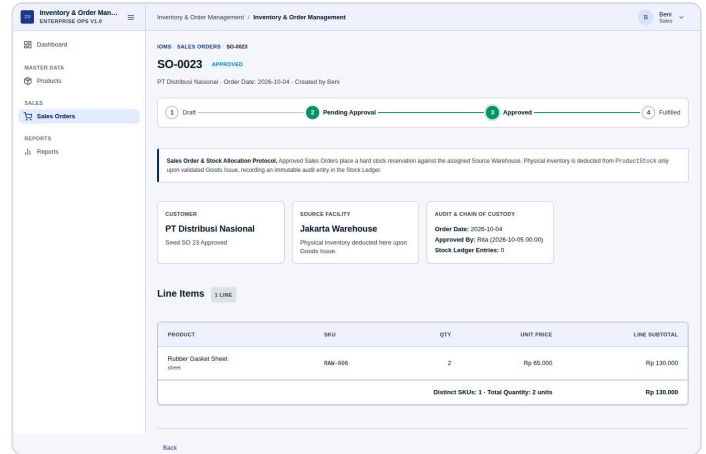
Validasi: simpan dengan qty 0



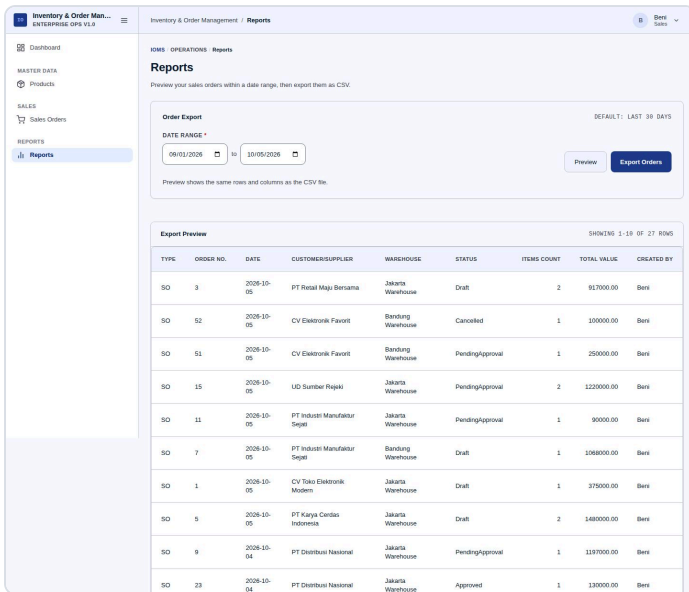
Setelah Submit: Pending Approval (tanpa Approve)



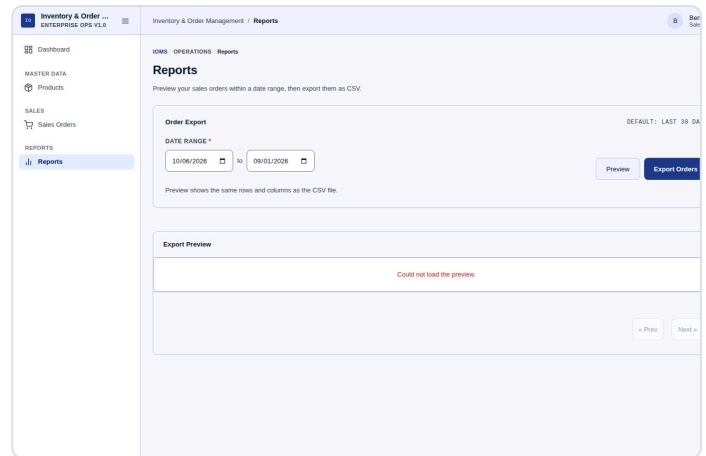
Konfirmasi Cancel Order



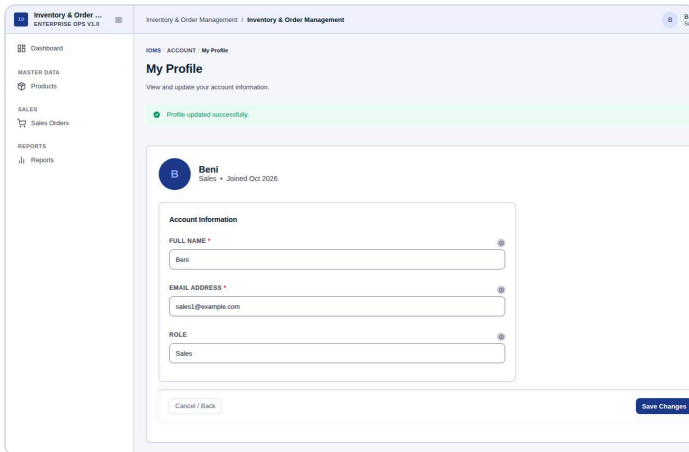
SO Approved: tidak ada Cancel/Issue



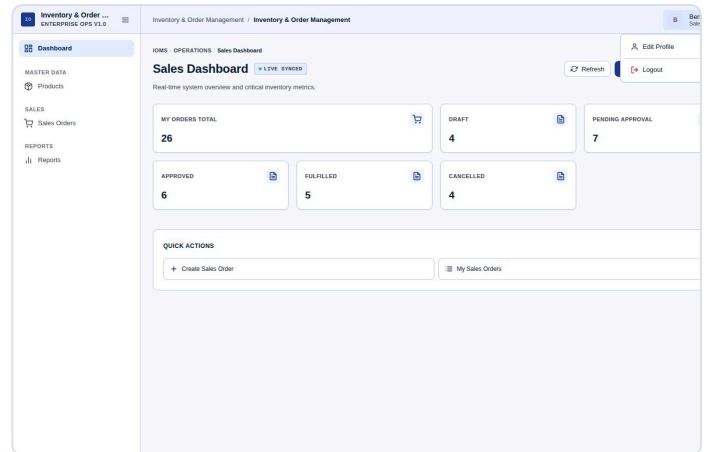
Reports: preview Export Orders (milik sendiri)



Tanggal mulai > akhir ditolak



Profil tersimpan



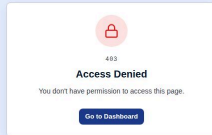
Menu akun / Logout

D.3 Pemeriksaan negatif (harus ditolak)

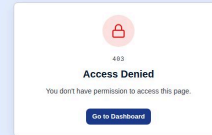
Coba	Hasil yang benar
Ketik <code>/users</code> , <code>/purchase-orders</code> , <code>/stock-ledger</code>	403
Buka detail SO milik Sales lain (<code>/sales-orders/{id}</code>)	403
POST langsung ke <code>/sales-orders/{id}/approve</code> atau <code>/reject</code> , termasuk untuk SO sendiri	Ditolak: "Only an administrator can approve a sales order."
POST ke <code>/sales-orders/{id}/issue</code>	403
Export Stock Ledger (<code>/reports/export/stock-ledger</code>)	403

Buka /categories , /warehouses , /suppliers

Tampil read-only (HTTP 200, tanpa tombol kelola) — halaman daftar hanya butuh login; aksi create/edit/deactivate tetap 403.



403 untuk /users, /purchase-orders, /stock-ledger, /products/create (halaman 403 yang sama)



SO milik Sales lain: 403

Inventory & Order Management / Inventory & Order Management

Categories

Organize product catalog classifications, manage category codes, and monitor SKU distributions.

TOTAL CATEGORIES
4
Registered in the system

ACTIVE CATEGORIES
4
Available for new products

TOTAL ASSIGNED SKUs
100 SKU
Linked to registered products

UNASSIGNED / EMPTY
0
SKU linked (can be deleted)

CATEGORY NAME

CATEGORY CODE

STATUS
All Status

SORT
Name (A-Z)

Search

Reset

Export CSV

CODE	CATEGORY NAME & DESCRIPTION	ASSIGNED SKUs	STATUS	LAST UPDATED
CAT-ELEC-01	Electronics Electronic devices and components	23 SKU	ACTIVE	06/10/2024 00:49
CAT-OFFC-01	Office Supplies Stationery and general office consumables	26 SKU	ACTIVE	06/10/2024 00:49

Press **Meta+Shift+L**

/categories lewat URL: tampil read-only

Inventory & Order Management / Inventory & Order Management

Customers

15 customers registered

NAME

CONTACT PERSON

PHONE

EMAIL

STATUS
Select status...

Search

Reset

NAME	CONTACT PERSON	PHONE	EMAIL	STATUS
CV Elektronik Favorit	Joko Susilo	024-6670123	order@elektronikfavorit.id	ACTIVE
CV Gudang Kita	Taufik Ismail	022-6666789	sales@gudangkita.id	ACTIVE
CV Toko Elektronik Modern	Fajar Nugroho	022-6662345	info@tokoelektronikmodern.co.id	ACTIVE
PT Distribusi Nasional	Yusuf Hidayat	031-6663456	order@distribusinasional.co.id	ACTIVE

/customers lewat URL: read-only

Permission Warehouse: `purchase_orders.manage`, `sales_orders.issue`, `stock_ledger.view`, `reports.stock_ledger.view`. Tidak punya submit/cancel PO, approve SO, `sales_orders.menu`, `customers.view`.

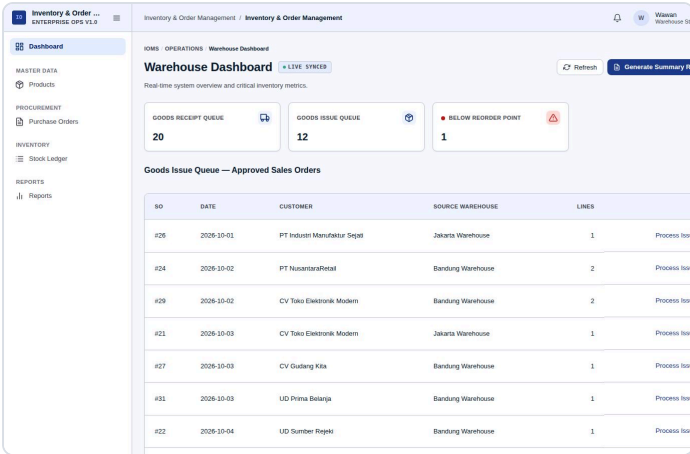
E.1 Peta menu & fungsinya

Menu (URL)	Untuk apa
Dashboard /dashboard	Antrean kerja gudang: Goods Receipt Queue (PO Ordered + PartiallyReceived, tombol Receive), Goods Issue Queue (SO Approved, tombol Process Issue; tertua dulu, 10 teratas), jumlah & daftar produk di bawah reorder point, bel notifikasi stok rendah.
Products /products (read-only)	Cek stok per gudang untuk menentukan restock.
Purchase Orders /purchase-orders	Melihat PO, membuat PO usulan restock (Draft), dan menerima barang .
Stock Ledger /stock-ledger	Audit/rekonsiliasi pergerakan stok fisik vs sistem.
Reports /reports	Analitik inventori + Export Stock Ledger (CSV).
My Profile	Ubah nama & email sendiri.

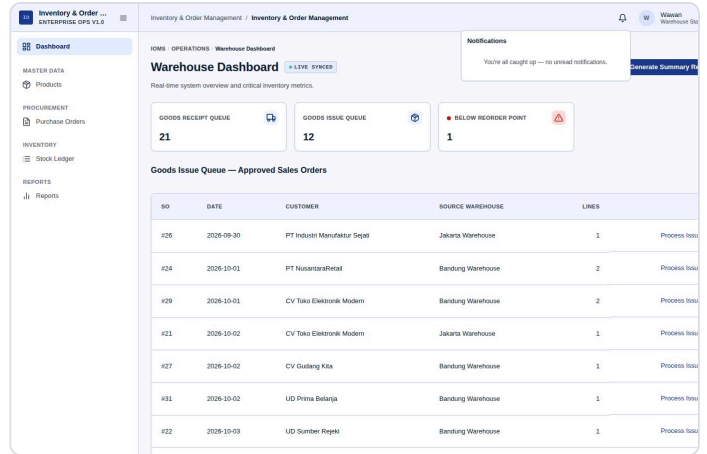
Catatan: tidak ada menu **Sales Orders** di sidebar. Jalur resmi untuk mengeluarkan barang adalah Dashboard → Process Issue → detail SO → **Issue Goods** (kodenya memang menjelaskan antrean dashboard sebagai satu-satunya jalur). Meski tanpa menu, `/sales-orders` dan detailnya tetap bisa dibuka lewat URL oleh Warehouse (hanya butuh login; Warehouse melihat semua SO).

E.2 Checklist how-to

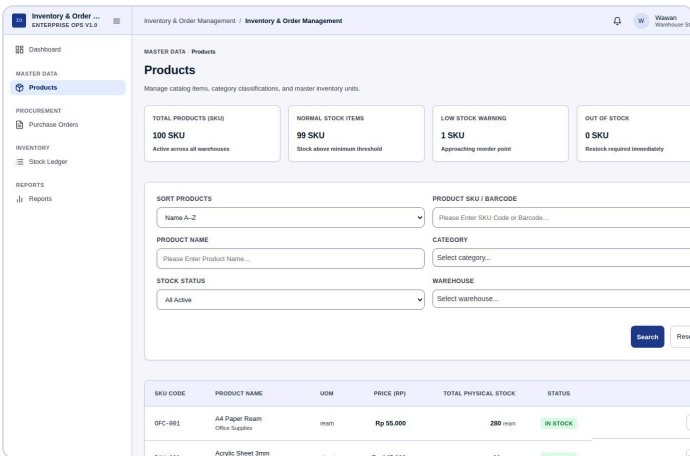
- ☐ Login `warehouse@example.com` → dashboard Warehouse; sidebar: Dashboard, Products, Purchase Orders, Stock Ledger, Reports; bel notifikasi tampil.
- ☐ Dashboard: angka Goods Receipt Queue & Goods Issue Queue cocok dengan jumlah PO Ordered/PartiallyReceived dan SO Approved; bila ada notifikasi, **Mark All as Read** bekerja.
- ☐ Products: lihat stok per gudang; tidak ada tombol Add/Edit.
- ☐ Purchase Orders: list + filter status/nomor/supplier; detail menampilkan progres penerimaan dan baris ledger terkait.
- ☐ Create PO (usulan restock) → tersimpan **Draft**; di detail Draft tidak ada tombol Submit Order dan tidak ada Cancel Order (Admin saja).
- ☐ PO Ordered/PartiallyReceived → **Receive Goods**: isi qty diterima per baris (bisa sebagian) → stok naik, baris Receipt di Stock Ledger, status PartiallyReceived/Received. Qty melebihi sisa → ditolak.
- ☐ Dashboard → **Process Issue** pada SO Approved → detail SO hanya menampilkan **Issue Goods** (tidak ada Approve/Reject) → stok turun, baris Issue di ledger, status Fulfilled.
- ☐ Issue dengan stok gudang asal kurang → ditolak "insufficient stock", stok & status tidak berubah.
- ☐ Stock Ledger: filter SKU/nama/jenis/gudang,urut kolom, 25 baris/halaman; baris Receipt & Issue yang baru dibuat muncul; tidak ada fitur edit/hapus.
- ☐ Reports: generate analitik; **Export Stock Ledger** → CSV. Tombol Export Orders: Warehouse tak punya izin ekspor SO/PO → 403 bila dipanggil langsung.
- ☐ My Profile: ubah nama/email → sukses. Logout.



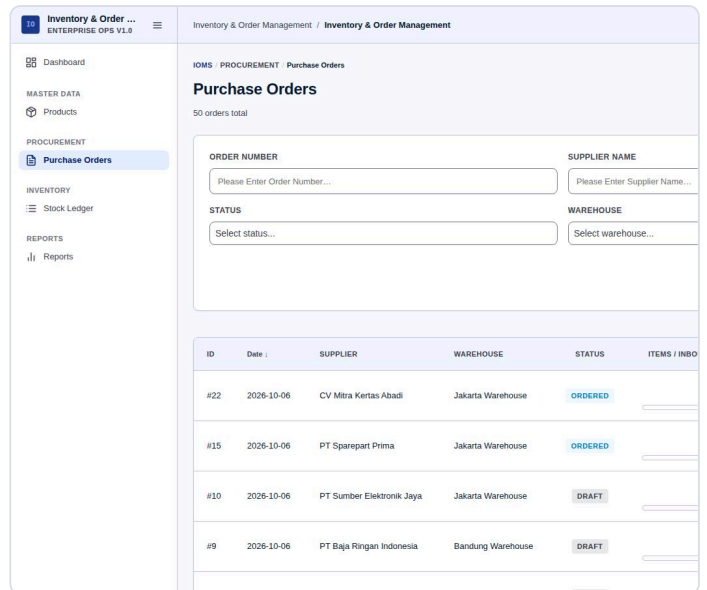
Dashboard Warehouse (antrean Receipt & Issue)



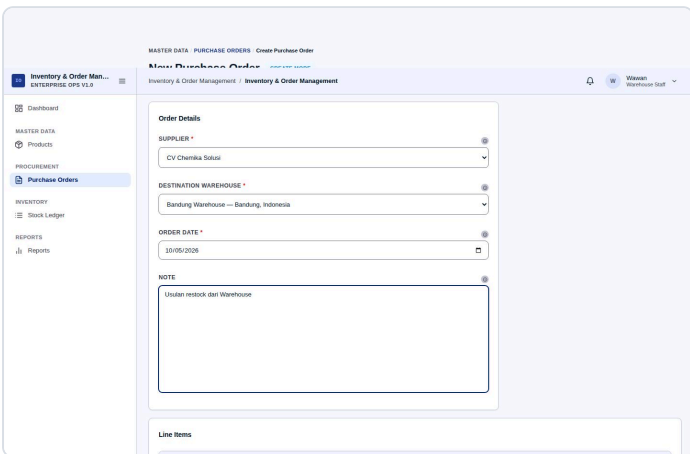
Bel notifikasi stok rendah



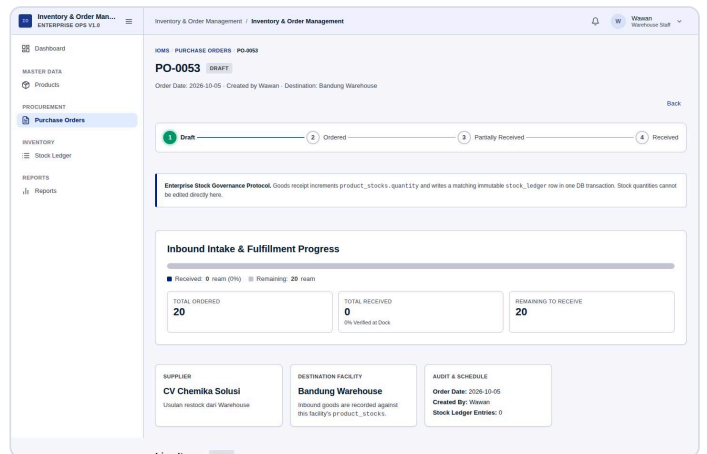
Products read-only



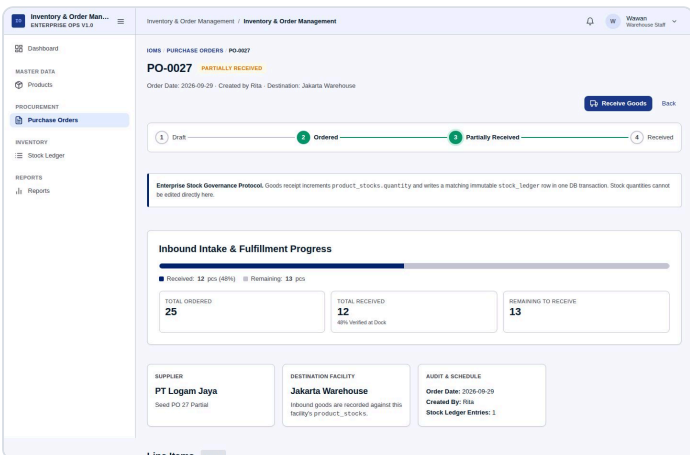
Daftar PO



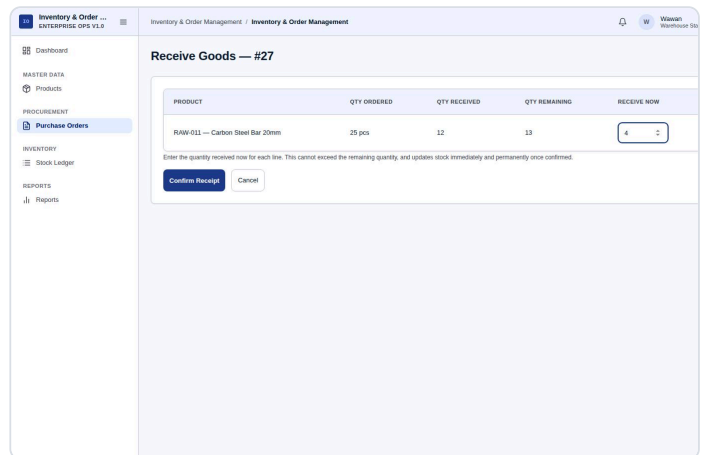
Create PO usulan



PO Draft: tanpa Submit/Cancel



PO Ordered: Receive Goods



Form Receive (isi sebagian)

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

Products

PROCUREMENT

Purchase Orders

INVENTORY

Stock Ledger

REPORTS

Reports

IOMS PURCHASE ORDERS PO-0027

PO-0027 **PARTIALLY RECEIVED**

Order Date: 2026-10-28 Created by Rita Destination: Jakarta Warehouse

1 Draft 2 Ordered 3 Partially Received 4 Received

Enterprise Stock Governance Protocol. Goods receipt increments product _stocks_ quantity and writes a matching immutable stock_ledger row in one DB transaction. Stock quantities cannot be edited directly here.

Inbound Intake & Fulfillment Progress

Received: 16 pcs (64%) Remaining: 9 pcs

TOTAL ORDERED: 25

TOTAL RECEIVED: 16 (with locked stock)

REMAINING TO RECEIVE: 9

SUPPLIER: PT Logam Jaya Sent PO 27 Partial

DESTINATION FACILITY: Jakarta Warehouse Inbound goods are recorded against this facility's product _stocks_.

AUDIT & SCHEDULE: Order Date: 2026-09-28 Created By: Rita Stock Ledger Entries: 2

Line Items

PartiallyReceived + audit log

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

Products

PROCUREMENT

Purchase Orders

INVENTORY

Stock Ledger

REPORTS

Reports

IOMS SALES ORDERS SO-0026

SO-0026 **APPROVED**

PT Industri Manufaktur Sejaht Order Date: 2026-10-01 Created by Grace

1 Draft 2 Pending Approval 3 Approved 4 Fulfilled

Sales Order & Stock Allocation Protocol. Approved Sales Orders place a held stock reservation against the assigned Source Warehouse. Physical inventory is deducted from ProductStock only upon validated Goods Issue, recording an immutable audit entry in the Stock Ledger.

CUSTOMER: PT Industri Manufaktur Sejaht Sent SO 26 Approved

SOURCE FACILITY: Jakarta Warehouse Physical inventory deducted here upon Goods Issue.

AUDIT & CHAIN OF CUSTODY: Order Date: 2026-10-01 Approved By: Rita (2026-10-02 00:00) Stock Ledger Entries: 0

Line Items

PRODUCT	SKU	QTY	UNIT PRICE	LINE SUBTOTAL
HDPE Sheet 5mm sheet	RAW-810	10	Rp 180.000	Rp 1.800.000
Distinct SKUs: 1 - Total Quantity: 10 units				Rp 1.800.000

Line Items

SO Approved: hanya Issue Goods

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

Products

PROCUREMENT

Purchase Orders

INVENTORY

Stock Ledger

REPORTS

Reports

IOMS SALES ORDERS SO-0028

SO-0028 **FULFILLED**

PT Industri Manufaktur Sejaht Order Date: 2026-10-01 Created by Grace

1 Draft 2 Pending Approval 3 Approved 4 Fulfilled

Sales Order & Stock Allocation Protocol. Approved Sales Orders place a held stock reservation against the assigned Source Warehouse. Physical inventory is deducted from ProductStock only upon validated Goods Issue, recording an immutable audit entry in the Stock Ledger.

CUSTOMER: PT Industri Manufaktur Sejaht Sent SO 26 Approved

SOURCE FACILITY: Jakarta Warehouse Physical inventory deducted here upon Goods Issue.

AUDIT & CHAIN OF CUSTODY: Order Date: 2026-10-01 Approved By: Rita (2026-10-02 00:00) Issued By: Vivian (2026-10-06 00:00) Stock Ledger Entries: 1

Line Items

PRODUCT	SKU	QTY	UNIT PRICE	LINE SUBTOTAL
HDPE Sheet 5mm sheet	RAW-810	10	Rp 180.000	Rp 1.800.000
Distinct SKUs: 1 - Total Quantity: 10 units				Rp 1.800.000

Audit Log

Setelah Issue: Fulfilled + ledger

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

Products

PROCUREMENT

Purchase Orders

INVENTORY

Stock Ledger

REPORTS

Reports

IOMS SALES ORDERS SO-0029

SO-0029 **APPROVED**

CV Toko Elektronik Modern Order Date: 2026-10-01 Created by Benji

1 Draft 2 Pending Approval 3 Approved 4 Fulfilled

Sales Order & Stock Allocation Protocol. Approved Sales Orders place a held stock reservation against the assigned Source Warehouse. Physical inventory is deducted from ProductStock only upon validated Goods Issue, recording an immutable audit entry in the Stock Ledger.

CUSTOMER: CV Toko Elektronik Modern Sent SO 29 Approved

SOURCE FACILITY: Bandung Warehouse Physical inventory deducted here upon Goods Issue.

AUDIT & CHAIN OF CUSTODY: Order Date: 2026-10-01 Approved By: Rita (2026-10-02 00:00) Stock Ledger Entries: 0

Line Items

PRODUCT	SKU	QTY	UNIT PRICE	LINE SUBTOTAL
Silicone Rubber Sheet	RAW-814	8	Rp 115.000	Rp 920.000
Copper Pipe 15mm psi	RAW-815	6	Rp 115.000	Rp 690.000
Distinct SKUs: 2 - Total Quantity: 14 units				Rp 1.610.000

Line Items

SO Approved dengan stok kurang

Bad Request

Insufficient stock: there is not enough stock on hand to issue this sales order.

Go to Dashboard

Issue ditolak: stok tidak mencukupi

Inventory & Order Management / Inventory & Order Management

Dashboard

MASTER DATA

Products

PROCUREMENT

Purchase Orders

INVENTORY

Stock Ledger

REPORTS

Reports

IOMS INVENTORY Stock Ledger

Stock Ledger

241 movements recorded

PRODUCT SKU: Please Enter Product SKU Code...

PRODUCT NAME: Please Enter Product Name...

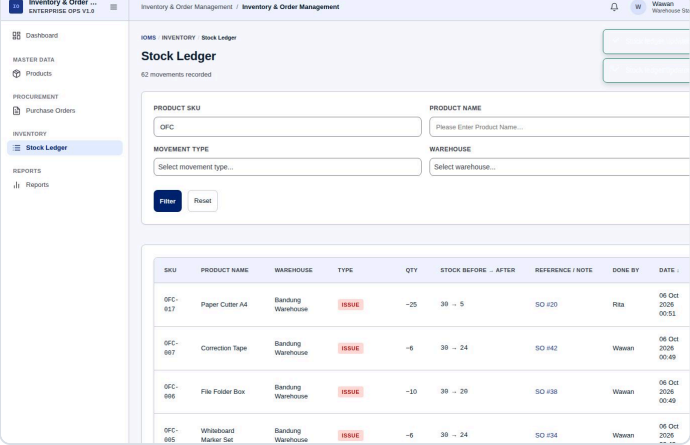
MOVEMENT TYPE: Select movement type...

WAREHOUSE: Select warehouse...

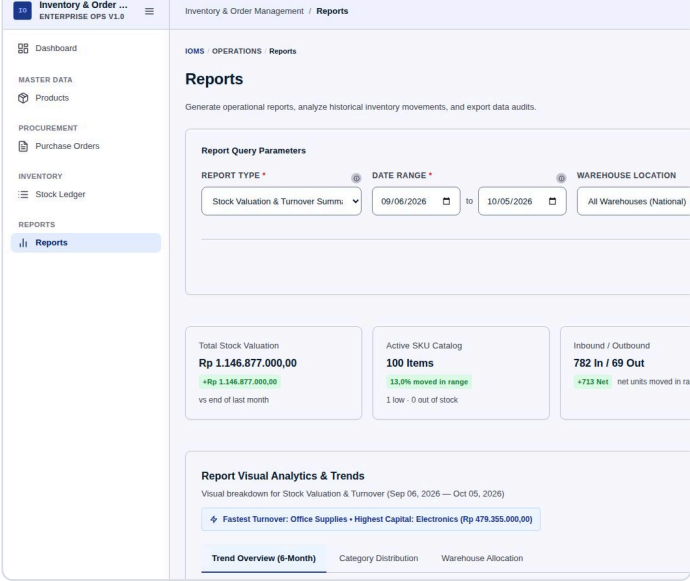
Filter Reset

SKU	PRODUCT NAME	WAREHOUSE	TYPE	QTY	STOCK BEFORE	AFTER
RAW-811	Carbon Steel Bar 20mm	Jakarta Warehouse	RECEIPT	+4	62	66
RAW-810	HDPE Sheet 5mm	Jakarta Warehouse	ISSUE	-10	65	55
OFC-897	Correction Tape	Bandung Warehouse	ISSUE	-6	36	24
ELEC-888	LED Monitor 24in	Jakarta Warehouse	ISSUE	-2	58	48

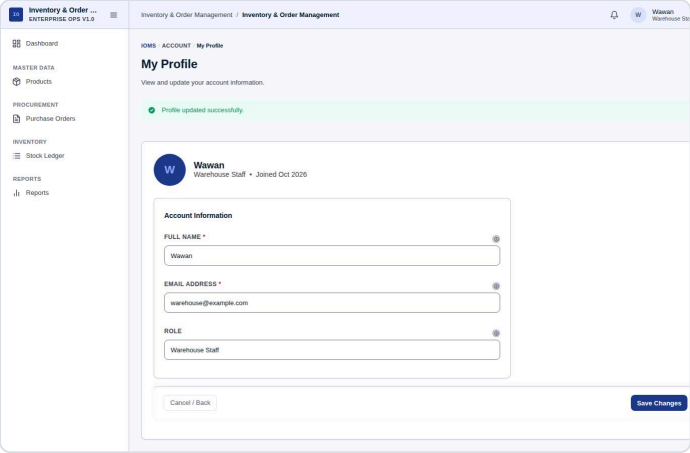
Stock Ledger



Filter SKU "OFC"



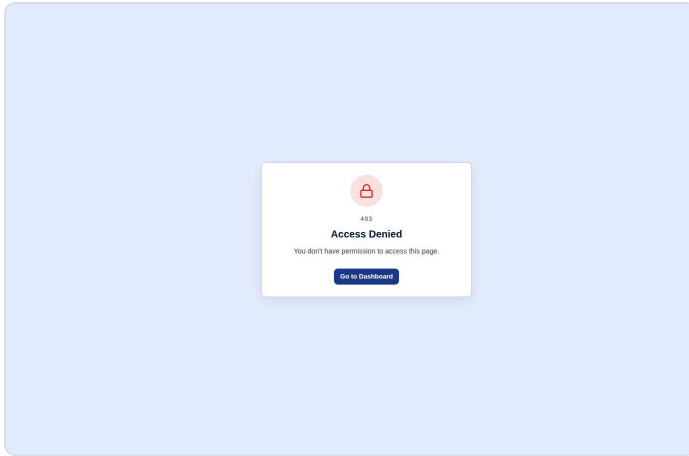
Reports + Export Stock Ledger



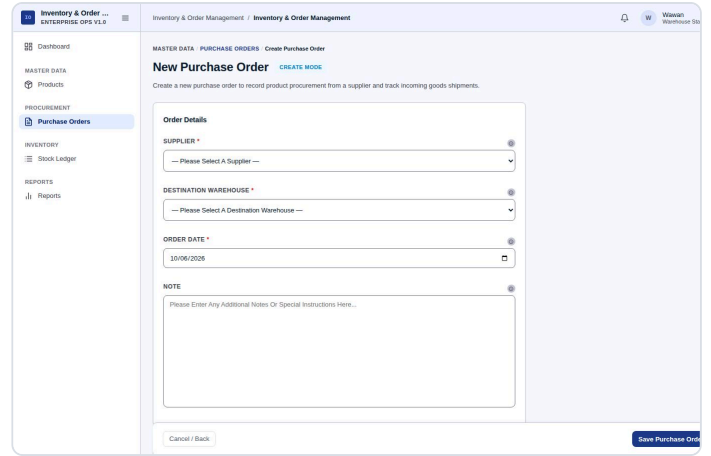
Profil tersimpan

E.3 Pemeriksaan negatif (harus ditolak)

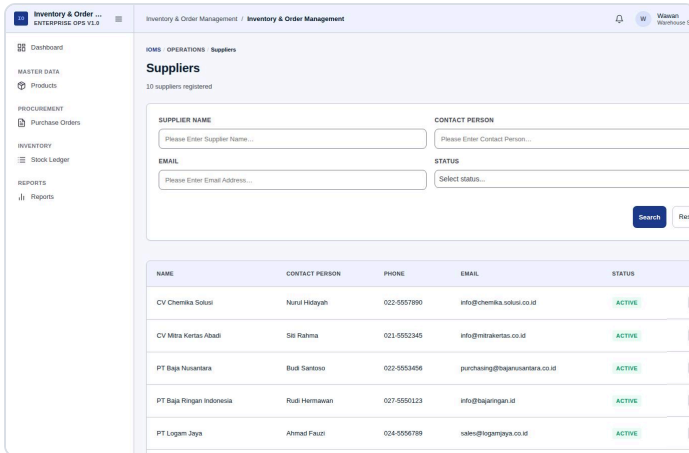
Coba	Hasil yang benar
Ketik <code>/users</code>	403
Ketik <code>/customers</code>	403 (butuh <code>customers.view</code>)
Buka <code>/sales-orders/create</code> atau POST buat SO	403 (butuh <code>sales_orders.menu</code>)
POST <code>/sales-orders/{id}/approve</code> atau <code>/reject</code>	403
POST <code>/purchase-orders/{id}/submit</code> atau <code>/cancel</code>	403
Create/edit/deactivate produk, kategori, gudang, supplier, customer	403
Buka <code>/sales-dashboard</code>	403



403 untuk /users, /customers, /sales-orders/create, /sales-dashboard (halaman 403 yang sama)



Pembanding: /purchase-orders/create boleh dibuka Warehouse



/suppliers lewat URL: read-only

F.1 Matriks permission (dari `database/seed.sql`)

Permission	Admin	Sales	Warehouse
users.manage, products/categories/warehouses/suppliers/customers.manage, master_data.menu	✓	✗	✗
customers.view	✓	✓	✗
purchase_orders.manage (lihat, buat, receive)	✓	✗	✓
purchase_orders.submit / .cancel	✓	✗	✗
sales_orders.menu (lihat menu, buat/edit SO)	✓	✓	✗
sales_orders.approve	✓	✗	✗
sales_orders.issue	✓	✗	✓
stock_ledger.view, reports.stock_ledger.view	✓	✗	✓
reports.sales_orders.view	✓	✓ (milik sendiri)	✗
reports.purchase_orders.view	✓	✗	✗

F.2 Koreksi yang dilakukan pada panduan ini

Versi lama	Fakta terverifikasi
Dashboard sebagian di-cache Memcached	Dashboard tidak di-cache; Memcached hanya untuk <code>product:<SKU></code> dan aturan validasi file; izin role di Redis (Lampiran B).
Menu Sales: "Sales Orders, Customers, Reports"	Sidebar Sales: Dashboard, Products, Sales Orders, Reports. Customers tidak ada di sidebar (hanya via URL).
Menu Warehouse: "Purchase Orders, Sales Orders (untuk issue), Stock Ledger"	Tidak ada menu Sales Orders. Issue dilakukan lewat Goods Issue Queue di dashboard. Sidebar juga memuat Products dan Reports.
Admin menu: "Master Data, Procurement, Sales, Inventory, Reports, Users"	Lengkap: ditambah Dashboard; Master Data berisi Products, Categories, Warehouses, Suppliers, Customers; Users ada di grup Administration.
Tombol "Terima", "Issue"	Label di aplikasi: "Receive Goods", "Issue Goods" (juga "Process Issue"/"Receive" di antrean dashboard).
Kartu Stok "tidak bisa diubah"	Tidak ada rute/fitur edit atau hapus baris ledger. Skema tidak punya trigger DB yang melarangnya — jadi perlindungan ada di tingkat aplikasi.
Sesi "bila Redis mati login tetap jalan"	Benar, dengan batasan: sesi lama tidak ikut pindah (Lampiran A.2).

F.3 Catatan jujur untuk tanya-jawab

- Halaman daftar Categories, Warehouses, Suppliers, dan Products hanya butuh login; Sales & Warehouse yang mengetik URL-nya melihat data read-only (tanpa tombol kelola). Yang dijaga ketat di server adalah aksi tulis. `/customers` dijaga `customers.view` (Warehouse → 403).
- Menu disembunyikan lewat `PermissionService` (yang di-cache 1 jam di Redis); guard server memakai data izin yang sama. Perubahan izin di database dapat tertunda hingga TTL habis.

- Menandai **inferensi** = disimpulkan dari pembacaan kode, belum dijalankan end-to-end.